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Refrigerator

Abstract

A refrigerator includes a cabinet defining a storage space, a door configured to open and close the storage space, a gap adjusting member configured to be provided between the door and the cabinet and to adjust a gap between the door and the cabinet, and a seating part configured to protrude from the door and to support the gap adjusting member. The gap adjusting member includes a contact part exposed to the rear of the door and being in contact with the cabinet in a state where the door is closed, and a plurality of adjusting parts protruding from the contact part toward the seating part at different heights from each other. The gap between the door and the cabinet is adjusted by selecting a height of the adjusting part which is in contact with the seating part by rotating the gap adjusting member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 17/348,213, filed on Jun. 15, 2021, which claims priority under 35 U.S.C. 119 and 35 U.S.C. 365 to Korean Patent Application No. 10-2020-0073938, filed on Jun. 17, 2020, the disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND

- (1) The present disclosure relates to a refrigerator.
- (2) In general, refrigerators are home appliances that allow low-temperature storage of food in an internal storage space that is shielded by a refrigerator door and are configured to store stored foods in an optimal condition by using cold air generated through heat exchange with refrigerant circulating in the refrigeration cycle to cool the interior of the storage space.
- (3) As such, refrigerators are gradually becoming larger and multifunctional in accordance with changes in dietary life and high-end products, and refrigerators having various structures and convenience devices in consideration of user convenience are being released.
- (4) In addition, in a case where a plurality of doors are disposed side by side in one refrigerator, or in a case where a plurality of refrigerators are disposed in a row when the heights in the front and rear directions between the doors do not match, the outer appearance thereof is not good and there is a problem that may cause user complaints.
- (5) In order to prevent this problem, in Korean Patent Laid-Open No. 10-2008-0047151, a refrigerator is disclosed which is provided with a step preventing means corresponding to the thickness of the gasket on the rear surface of the door or on the case, so that when the door is closed, a predetermined gap is maintained between the door and the case.
- (6) However, in such a conventional technique, if the step preventing means of the gasket is equal to the height of the gasket, the gasket cannot be in close contact with the interior of the refrigerator due to deformation of the gasket or the like during long-term use, and thus there is a problem that the interior of the refrigerator cannot be completely airtight, and if the step preventing means of the gasket is lower than the height of the gasket, there may also be a problem that the door does not close to the same height.
- (7) In particular, the conventional step prevention means cannot adjust the height of the door in the front and rear direction, and only one fixed height is provided, and thus there is a problem that it cannot be effectively responded in situations where height adjusting of both sides of the door is required due to the difference in load in the refrigerator or the like.
- (8) In order to prevent such a problem, in Korean Patent Laid-Open No. 10-2019-0066721, a refrigerator is disclosed which has a gap adjusting member provided at the upper end of the door so that the user can adjust the protruding height of the door when the door is closed.
- (9) However, in such a conventional technique, in order to adjust the protruding height of the door by using the gap adjusting member, the cover must be mounted again after removing the cover and then manipulating the gap adjusting member to set the height, which is inconvenient to manipulate. In addition, problems such as loss of the detachable cover or separation of the gap adjusting member can be expected.

SUMMARY

- (10) An object of an embodiment of the present disclosure is to provide a refrigerator that easily adjusts a protruding height of a front surface of a refrigerator door to eliminate a step from a surrounding structure.
- (11) An object of an embodiment of the present disclosure is to provide a refrigerator capable of

intuitively identifying an adjusting height when adjusting a step of a refrigerator door.

(12) An object of an embodiment of the present disclosure is to provide a refrigerator capable of ensuring the reliability of a step adjusting manipulation of a refrigerator door.

(13) An object of an embodiment of the present disclosure is to provide a refrigerator having a simple structure for adjusting a step of a refrigerator door.

(14) In a refrigerator according to an embodiment of the present disclosure, a gap adjusting member mounted on the rear surface of the door and in contact with the front surface of the cabinet in a state where the door is closed includes a contact part which is in contact with the cabinet, and a plurality of adjusting parts disposed to rotate with respect to the center of the contact part and having different heights, and in which the height of the gap adjusting member is adjusted by rotating the gap adjusting member.

(15) In a refrigerator according to an embodiment of the present disclosure, a gap adjusting member provided on a rear surface of a door and in contact with the front surface of the cabinet when the door is closed includes a contact part which is in contact with the cabinet, an adjusting part disposed along the front circumference of the contact part and formed to be stepped at different heights, and a height indicating part formed on the front surface of the contact part and indicating a height of the adjusting part at a position corresponding to the adjusting part.

(16) A refrigerator according to an embodiment of the present disclosure includes a seating part formed on a rear surface of a door, and a gap adjusting member mounted on the seating part and in contact with the front surface of the cabinet when the door is closed, in which the gap adjusting member includes a contact part which protrudes outside the seating part to contact the cabinet, an adjusting part which is continuously disposed in a polygonal shape on a front surface of the contact part and has different heights, in which an inside of the seating part is formed in a polygonal shape corresponding to the adjusting part so that the adjusting part is inserted.

(17) A refrigerator according to an embodiment of the present disclosure includes a seating part formed on a rear surface of a door, and a gap adjusting member mounted to be movable in the front and rear direction to the seating part and in contact with the front surface of the cabinet when the door is closed, in which the gap adjusting member include a contact part in contact with the cabinet, an insertion protrusion extending from a contact part and mounted through the seating part and serving as a rotation shaft of the gap adjusting member, and an adjusting part disposed in front of the contact portion to rotate around an insertion protrusion and having different heights, in which the height of the gap adjusting member is adjusted by rotating the gap adjusting member after moving the gap adjusting member backward.

(18) A refrigerator according to an embodiment of the present disclosure may include a cabinet configured to form a storage space, a door configured to open and close the storage space, a gap adjusting member configured to be provided between the door and the cabinet and to adjust a gap between the door and the cabinet, and a seating part configured to protrude from the door and to support the gap adjusting member, in which the gap adjusting member may include a contact part exposed to the rear of the door and being in contact with the cabinet in a state where the door is closed, and a plurality of adjusting parts protruding from the contact part toward the seating part at different heights from each other, and in which the gap between the door and the cabinet may be adjusted by selecting a height of the adjusting part which is in contact with the seating part by rotating the gap adjusting member.

(19) The plurality of adjusting parts may be continuously disposed in a height order.

(20) The plurality of adjusting parts may be located at equal gaps with respect to the center of the contact part.

(21) The plurality of adjusting parts may be disposed to rotate with respect to the center of the contact part.

(22) The adjusting part may be sequentially formed to be stepped in a rotation direction.

(23) The contact part may be formed in a polygonal shape, and the adjusting part may be formed

along each side of the contact part.

(24) The adjusting part may be formed to be stepped stepwise by the same height difference in a rotation direction along each side of the contact part.

(25) The adjusting part may include a main adjusting part protruding along each side of the contact part, and a sub adjusting part protruding from the inside of the main adjusting part more than the main adjusting part and formed along each side of the main adjusting part, and in which the height difference between each side of the plurality of main adjusting part and the plurality of sub adjusting part may be formed to be same.

(26) The main adjusting part and the sub adjusting part may be sequentially formed to be stepped in a state of being rotated 180° with respect to the center of the contact part.

(27) The sub adjusting part may be formed to have the same cross-sectional size as the inside of the seating part and may be inserted into the seating part.

(28) The seating part may include a main support part forming a circumferential surface of the seating part and supporting the main adjusting part, and a sub support part which is formed on an inside of each side of the main support part and includes a sub-support part for supporting the sub-adjusting part, and in which the main support part and the sub support part may be formed to be stepped with the same height difference as the main adjusting part and the sub adjusting part, respectively.

(29) The seating part may be opened toward the cabinet, and at least a portion of the adjusting part may be inserted.

(30) The seating part may include a main support part that forms a circumferential surface of the seating part and is in contact with the adjusting part.

(31) The seating part may include a sub support part formed in an opened inside of the seating part and is in contact with another part of the adjusting part inserted into the seating part.

(32) The seating part may be formed in a polygonal shape corresponding to the adjusting part, and the sub support part may be configured with a plurality of protrusions protruding from the inside surface of each side of the main support part.

(33) The contact part may be formed with an insertion protrusion inserted into the seating part, and the adjusting part may be disposed to rotate about the insertion protrusion.

(34) A through-hole through which the protrusion passes may be formed in the seating part, and an elastic part may be formed in the insertion protrusion to protrude more than a diameter of the through-hole and to maintain a state where the insertion protrusion is mounted on the seating part.

(35) The gap adjusting member may be at least partially formed of a material that is elastically deformable.

(36) An upper cap deco forming a portion of the upper and rear surfaces of the door may be provided at the upper end of the door, and the seating part may be formed in the upper cap deco.

(37) A top cover which protrudes upward and forms a surface facing the door may be formed on the upper surface of the cabinet, and the contact part may be in contact with the top cover in a state where the door is closed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a state where a refrigerator according to an embodiment of the present disclosure is installed.

(2) FIG. 2 is a perspective view illustrating the refrigerator.

(3) FIG. 3 is a perspective view illustrating a refrigerator door according to an embodiment of the present disclosure viewed from the rear.

(4) FIG. 4 is a partial side view illustrating a state of a gap adjusting member in a state where the

refrigerator door is closed.

(5) FIG. 5 is an exploded perspective view illustrating a coupling relationship between the gap adjusting member and the seating part.

(6) FIG. 6 is a front view illustrating a seating part according to an embodiment of the present disclosure.

(7) FIG. 7 is a perspective view illustrating the gap adjusting member viewed from one side.

(8) FIG. 8 is a perspective view illustrating the gap adjusting member viewed from another side.

(9) FIG. 9 is a cut-away perspective view illustrating a state where the gap adjusting member is seated.

(10) FIG. 10 is a perspective view illustrating a state where the gap adjusting member is pulled out for height adjustment.

(11) FIG. 11 is a cut-away perspective view of FIG. 9.

(12) FIG. 12 is a cut-away perspective view illustrating the height adjusting state of a first level height adjusting state of the gap adjusting member.

(13) FIG. 13 is a cut-away perspective view illustrating a second level height adjusting state of the gap adjusting member.

(14) FIG. 14 is a cut-away perspective view illustrating a third level height adjusting state of the gap adjusting member.

(15) FIG. 15 is a cut-away perspective view illustrating a fourth level height adjusting state of the gap adjusting member.

(16) FIG. 16 is a view schematically illustrating a state of adjusting a step between the door and an adjacent configuration.

(17) FIG. 17 is a perspective view illustrating a refrigerator door according to another embodiment of the present disclosure viewed from the rear.

(18) FIG. 18 is an exploded perspective view illustrating a coupling relationship between the gap adjusting member and the seating part.

(19) FIG. 19 is a cut-away perspective view taken along line XIX-XIX' of FIG. 17.

(20) FIG. 20 is a cut-away perspective view of a state where the gap adjusting member is pulled out for height adjustment in FIG. 15.

(21) FIG. 21 is a partial perspective view illustrating a state where a gap adjusting member is mounted according to another embodiment of the present disclosure.

(22) FIG. 22 is an exploded perspective view illustrating a state where the gap adjusting member of the present disclosure is separated from the seating part.

(23) FIG. 23 is a cut-away perspective view taken along line XXIII-XXIII' of FIG. 21.

(24) FIG. 24 is a perspective view illustrating a refrigerator door according to another embodiment of the present disclosure viewed from the rear.

(25) FIG. 25 is a view schematically illustrating a state of adjusting a step between the refrigerator doors.

(26) FIG. 26 is an exploded perspective view illustrating a state where a refrigerator door according to another embodiment of the present disclosure is separated.

(27) FIG. 27 is a perspective view illustrating a refrigerator according to another embodiment of the present disclosure.

(28) FIG. 28 is a partial perspective view illustrating an upper portion of the cabinet of the refrigerator.

(29) FIG. 29 is a rear view illustrating a refrigerator door according to another embodiment of the present disclosure.

(30) FIG. 30 is a perspective view illustrating a gap adjusting member according to another embodiment of the present disclosure.

(31) FIG. 31 is a cut-away perspective view taken along line XXXI-XXXI' of FIG. 29.

(32) FIG. 32 is a cut-away perspective view illustrating a manipulation state of the step adjustment

member.

(33) FIG. 33 is a partial perspective view illustrating a state where the door of the refrigerator is closed.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(34) Hereinafter, specific embodiments of the present disclosure will be described in detail together with the drawings. However, the present disclosure cannot be said to be limited to the embodiments in which the spirit of the present disclosure is presented, and other disclosures that are regressive by addition, change, deletion, or the like of other components, or other embodiments included within the spirit scope of the present disclosure can be easily suggested.

(35) Define the direction prior to the explanation thereof. According to an embodiment of the present disclosure, the direction toward the front surface of the door as illustrated in FIG. 2 can be defined as front, the direction toward the cabinet with respect to the front surface of the door can be defined as rear, and the direction toward the floor where the refrigerator is installed can be defined as downward, and the direction away from the floor can be defined as upward.

(36) FIG. 1 is a perspective view illustrating a state where a refrigerator according to an embodiment of the present disclosure is installed. In addition, FIG. 2 is a perspective view illustrating the refrigerator.

(37) As illustrated in the drawings, the refrigerator 1 according to the embodiment of the present disclosure has an overall outer appearance by a cabinet 10 forming a storage space with an opened front surface and doors 20 and 30 opening and closing the storage space.

(38) In addition, the refrigerator 1 may be mounted so as to harmonize with the furniture or wall O of the indoor space. For example, as illustrated in FIG. 1, the refrigerator 1 may be installed in an indoor space such as a kitchen and may be disposed adjacent to furniture or a wall O to harmonize. In other words, a space corresponding to the size of the refrigerator 1 may be provided in the furniture or wall O, and the refrigerator 1 may be received or may be disposed in a built-in type therein.

(39) In addition, other refrigerators 1' may be continuously arranged in parallel on the side of the refrigerator 1. In other words, a space in which a plurality of refrigerators 1 and 1' can be disposed may be provided by furniture or a wall O.

(40) The plurality of refrigerators 1 and 1' may have the same structure, and refrigerators 1 and 1' having various structures may be disposed in combination as needed. For example, as illustrated in FIG. 1, in the refrigerator 1 disposed on the right side, doors 20, 31, and 32 may be disposed vertically. In other words, the refrigerator 1 on the right may be configured such that a door 20 that is opened and closed in a rotating manner is provided at the upper side, and doors 31 and 32 that are opened and closed in a sliding manner are provided at the lower side.

(41) In addition, a refrigerator 1' may be further provided on the left side of the refrigerator 1 on the right side. The refrigerator 1' disposed side by side on the left may be configured such that an upper storage space is opened and closed by a pair of doors 33 and 34, and a lower storage space is opened and closed by a pair of doors 35. In addition, the doors 33, 34, and 35 may be configured to open and close in a rotating manner, and these doors 33, 34, and 35 may be referred to as French type doors.

(42) Meanwhile, front plates 21 having the same outer appearance may be mounted on all the front surfaces of the doors 20, 31, 32, 33, 34, and 35 of the refrigerator 1 on the right and the refrigerator 1' on the left. Accordingly, when viewed from the front, the overall outer appearance of the refrigerators 1 and 1' may be formed by the plurality of front plates. In this case, the front plates of the refrigerators 1 and 1' may have the same structure but only differ in size.

(43) In addition, the refrigerator 1 is disposed adjacent to the adjacent refrigerator 1', furniture, or wall O, and adjacent refrigerator 1', furniture, or walls O within a range that does not interfere when the doors 20 and 30 are opened or closed can be disposed close to minimize the clearance therebetween.

(44) Meanwhile, in a state where the doors **20**, **31**, **32**, **33**, **34** and **35** of the refrigerators **1** and **1'** are closed, it is preferable that the front surfaces of the refrigerators **1** and **1'** are located on the same plane, but according to a load applied to the door **20**, **31**, **32**, **33**, **34**, and **35** during use of the refrigerators **1**, **1'** or a state of the installation and use of the refrigerators **1**, **1'**, the closing amount of the door **20**, **31**, **32**, **33**, **34**, and **35** may vary, and the protrusion degree of the doors **20**, **31**, **32**, **33**, **34**, and **35**, that is, the height in the front and rear direction may vary.

(45) In this case, the front surfaces of some of the doors **20**, **31**, **32**, **33**, **34**, and **35** of the doors **20**, **31**, **32**, **33**, **34**, and **35** further protrude and are located in front of the front surfaces of other doors **20**, **31**, **32**, **33**, **34**, and **35** and thus the doors **20**, **31**, **32**, **33**, **34**, and **35** have a step with respect to each other and do not have a sense of unity with other doors **20**, **31**, **32**, **33**, **34**, and **35** or surrounding furniture or walls **O**. Accordingly, the user can adjust the protruding height of the doors **20**, **31**, **32**, **33**, **34**, and **35** using the gap adjusting member **40**, which will be described below.

(46) The gap adjusting member **40** can be applied to all of the doors **20**, **31**, **32**, **33**, **34** and **35** of the refrigerator **1** and **1'** and in the following, for the convenience of understanding and explanation, and to prevent duplication of description, the refrigerator **1** on the right side of the refrigerators **1** and **1'** of FIG. **1** will be described in more detail.

(47) The refrigerator **1** may have an outer appearance formed by the cabinet **10** and the doors **20**, **31**, and **32**. In addition, in the cabinet **10**, the storage space may be divided vertically, an upper door **20** may be provided in the upper storage space, and lower doors **31** and **32** may be provided in the lower storage space.

(48) In addition, a top cover **11** may be provided on the upper surface of the cabinet **10**. The top cover **11** may extend along the front upper end of the cabinet **10** and may extend to a height corresponding to the height of the upper door **20**. Therefore, when the upper door **20** is closed, the refrigerator can have a more united outer appearance. In addition, the top cover **11** may receive a part of the upper hinge **201** for rotation of the upper door **20**. In addition, the top cover **11** may be provided with a display and a manipulation part for displaying the operating state of the refrigerator **1**.

(49) An upper door **20** may be provided in the upper storage space, and a lower door **30** may be provided in the lower storage space. The upper door **20** may open and close the upper storage space by rotation. An upper hinge **201** and a lower hinge **202** are provided at the upper and lower ends of one side of the upper door **20**, respectively, and the upper door **20** may be rotatably mounted on the cabinet **10** by the upper hinge **201** and the lower hinge **202**. Accordingly, the upper door **20** may be referred to as a rotary door.

(50) In addition, the lower door **30** may be pushed in and pulled out in a drawer type to open and close the lower storage space, and the lower door **30** may be referred to as a drawer door.

(51) The lower door **30** may be provided with two upper and lower doors, and the lower storage space opened and closed by the lower door **30** can consist of one space or also can be partitioned as a space in which each of the lower doors **30** is received.

(52) Meanwhile, the front outer appearances of the upper door **20** and the lower door **31** and **32** may be formed by the front plate **21**. The front plates **21** provided in the upper door **20** and the lower doors **31** and **32**, respectively, are different only in size and may have the same outer appearance.

(53) In addition, the gap adjusting member **40** for adjusting the upper door **20** and the lower door **31** and **32** to maintain the same front height in a state where the upper door **20** and the lower door **31** and **32** are closed, respectively is provided in the upper door **20** and the lower door **31** and **32**.

(54) Hereinafter, for convenience of explanation and understanding, the gap adjusting member **40** will be described based on the upper door **20**. In addition, hereinafter, the upper door **20** may be referred to as a door **20**.

(55) FIG. **3** is a perspective view illustrating a refrigerator door according to an embodiment of the present disclosure viewed from the rear. In addition, FIG. **4** is a partial side view illustrating the

state of the gap adjusting member when the refrigerator door is closed.

(56) As illustrated in the drawing, the outer appearance of the door **20** may be formed by a front plate **21** forming a front surface, a door liner **25** forming a rear surface, an upper cap deco **22** and a lower cap deco **23** forming the upper and lower surfaces. In addition, side frames **24** may be further included on the both left and right side surfaces of the door **20** to form the outer appearance of the both left and right side surfaces. In addition, in the interior of the door **20**, an insulating material is molded by a foaming liquid to insulate the storage space.

(57) In detail, the front plate **21** may be formed of a tempered glass or metal material to form the front outer appearance of the door **20**. When the side frame **24** is not provided on the door **20**, the front plate **21** may form the outer appearance of both side surfaces of the door **20**. In addition, the front plate **21** may be formed in a panel structure detachable from the door **20**, and it is possible to form an outer appearance that harmonizes with other home appliances such as the neighboring refrigerator **1** and furniture or wall **O**.

(58) The door liner **25** may be formed in a plate shape to form the outer appearance of the rear surface of the door **20** and may be formed of a plastic material. The door liner **25** may provide a structure in which a pocket or basket for storage is mounted as necessary. In addition, a gasket **26** may be mounted around the door liner **25**. When the door **20** is closed, the gasket **26** may be in close contact with the front surface of the cabinet **10** to seal the storage space.

(59) The upper cap deco **22** may be formed of a plastic material and may form the outer appearance of the upper surface and the rear surface **222** of the door **20** when mounted on the door **20**. In addition, the front surface of the upper cap deco **22** may support the upper end of the front plate **21**. In addition, both left and right side surfaces of the upper cap deco **22** may be coupled to an upper end of the side frame **24**.

(60) In particular, the rear surface **222** of the upper cap deco **22** may form the outer appearance of the rear upper surface of the door **20**. The rear surface of the door **20** above the gasket **26** may be formed by the upper cap deco **22**. The rear surface of the upper cap deco **22** may be formed in a flat shape and may form a surface facing the top cover **11** provided on the upper end of the cabinet **10** and/or the upper surface of the cabinet **10**. In addition, a hinge mounting part **221** on which the upper hinge **201** is mounted may be formed at one end of the upper cap deco **22**.

(61) In addition, the gap adjusting member **40** may be provided on the rear surface **222** of the upper cap deco **22**. The gap adjusting member **40** protrudes from the rear surface of the upper cap deco **22** and may contact the front upper end of the cabinet **10**.

(62) The gap adjusting member **40** may be located further outside the gasket **26** and may be located on one side farther than the rotation axis with respect to the center of the door **20**. In addition, as illustrated in FIG. **4**, when the door **20** is closed, at least a portion of the gap adjusting member **40** may be in contact with the front upper end of the cabinet **10**. Accordingly, the door **20** may be supported by the cabinet **10** and may serve as a stopper when the door **20** is closed. The gap adjusting member **40** may be configured to be in contact with the top cover **11**, and the top cover **11** may be omitted depending on the height of the cabinet **10**.

(63) Meanwhile, the gap adjusting member **40** can adjust the degree of protrusion from the rear surface of the door **20**, and by the manipulation of the gap adjusting member **40**, the gap **D** between the rear surface of the door and the front surface of the cabinet can be adjusted. In addition, through this, the degree of protrusion in the front of the door **20** can be adjusted, and other doors **31**, **32**, **33**, **34**, and **35** disposed adjacent to each other, or home appliances or furniture or wall **O**, and the front surface can be entirely adjusted evenly.

(64) In other words, when the door **20** is closed, in a case where the step is generated by the level difference between the other doors **31**, **32**, **33**, **34**, and **35**, home appliances, furniture, or wall **O** in the front and rear direction, the height of the front protrusion of the door **20** may be adjusted by manipulating the gap adjusting member **40**. Through the manipulation of the gap adjusting member **40**, the door **20** may be located on the same plane without a step with other neighboring doors **31**,

32, 33, 34, and 35, furniture, or wall **O** in a closed state. In this case, the same plane does not mean only exactly the same plane and may mean the degree to which the step is checked between the doors **20, 31, 32, 33, 34, and 35** or between the door **20, 31, 32, 33, 34, and 35** and furniture or wall **O** when viewed in a state of being spaced apart by the predetermined distance from the front of the refrigerator **1**.

(65) Hereinafter, the structure of the gap adjusting member **40** will be described in more detail with reference to the drawings.

(66) FIG. 5 is an exploded perspective view illustrating a coupling relationship between the gap adjusting member and the seating part. In addition, FIG. 6 is a front view illustrating the seating part according to an embodiment of the present disclosure.

(67) As illustrated in the drawing, a seating part **50** on which the gap adjusting member **40** is mounted may be formed on the rear surface of the door **20**. The seating part **50** may be formed on the rear surface of the upper cap deco **22**. The seating part **50** may be formed at a different position in the rear space of the door **20** outside the gasket **26** so as to be in contact with the cabinet **10** according to the structure of the door **20**.

(68) Hereinafter, an example is described that the seating part **50** is formed on the upper cap deco **22**, but the seating part **50** may be located at various positions on the rear surface of the door **20**.

(69) The seating part **50** may be formed to protrude from one side of the rear surface **222** of the upper cap deco **22** facing the front upper end of the cabinet **10**. The seating part **50** may be formed such that the rear surface thereof is opened, and a recessed part is formed inside the opened rear surface to form an receiving space **500** in which a part of the gap adjusting member **40** is received.

(70) The seating part **50** may be provided with a support part **53** on which the adjusting part **43** of the gap adjusting member **40** is seated. In addition, the height of the gap adjusting member **40** may be adjusted by a combination of a contact surface of the adjusting part **43** and the support part **53**.

(71) The seating part **50** may be formed to have a polygonal cross-sectional shape when viewed from the front. In addition, the seating part **50** is formed so that each surface of the polygonal shape has a different height and may support the gap adjusting member **40**. In an embodiment of the present disclosure, it is described for example that the shape of the seating part **50** and the adjusting part **43** are formed in a rectangular cross-section, but the cross-section of the seating part **50** can be formed in a polygonal shape such as triangle, pentagon, and hexagon, and accordingly, the disposition of the support part **53** will also be possible according to the cross-sectional shape of the seating part.

(72) For example, the seating part **50** may have a rectangular cross-sectional shape, and support parts **53** having different heights may be formed on four protruding surfaces. At least two or more surfaces of the support part **53** may contact the gap adjusting member **40** to determine a protruding height of the gap adjusting member **40**. Each of the support parts **53** facing each other may be configured to support the adjusting part **43** to stably support the gap adjusting member **40**.

(73) The support part **53** may include a main support part **51** forming a circumference of the seating part **50** and a sub support part **52** formed on an inner surface of the main support part **51**.

(74) In detail, referring to FIG. 6, the main support part **51** of the seating part **50** includes a first main support part **511** forming a left surface, a second main support part **512** forming an upper surface, and a third main support part **513** forming a right surface, and a fourth main support part **514** forming a lower surface.

(75) A seating part guide **515** for guiding the alignment and mounting of the gap adjusting member **40** may be formed on an outer surface of the seating part **50**. The seating part guide **515** may be formed on an outer surface of the second main support part **512** that forms an upper surface of the seating part **50**. The seating part guide **515** may protrude upward from an outer surface of the second main support part **512** and may protrude forward and backward along the center of the second main support part **512**.

(76) When the gap adjusting member **40** is mounted, the user may mount the gap adjusting member

40 based on the seating part guide **515**. In addition, when the gap adjusting member **40** is mounted, the adjusting member guide **415** formed on the gap adjusting member **40** may be aligned with the seating part guide **515**. Accordingly, the gap adjusting member **40** can be mounted at an accurate position, and the gap adjusting member **40** can be manipulated to an exact height desired by the user.

(77) Meanwhile, the first main support part **511**, the second main support part **512**, the third main support part **513**, and the fourth main support part **514** are sequentially connected to each other to have a rectangular cross-sectional shape and an receiving space **500** can be formed therein. The first main support part **511**, the second main support part **512**, the third main support part **513**, and the fourth main support part **514** may protrude vertically from the rear surface of the door **20** and can protrude to have different heights with each other.

(78) For example, the first main support part **511** may have a set height, and the second main support part **512**, the third main support part **513**, and the fourth main support part **514** are formed to gradually increase in order. In this case, the first main support part **511**, the second main support part **512**, the third main support part **513**, and the fourth main support part **514** may protrude to have a sequential height difference. For example, when the first main support part **511** is formed to have a height of 9 mm, the second main support part **512** is formed to have a height of 9.5 mm, the third main support part **513** is formed to have a height of 10 mm, and the fourth main support part **514** is formed to have a height of 10.5 mm.

(79) In other words, the height of the plurality of main support parts **51** can be adjusted by one level (0.5 mm) each time it is rotated by a 90° angle in the clockwise direction and can return to the original height if the plurality of main support parts rotate one turn (360°).

(80) Therefore, when the gap adjusting member **40** is manipulated in a state where the gap adjusting member **40** is rotated stepwise by a 90° angle at the position where the gap adjusting member **40** is initially mounted, the contact height of the main adjusting part **41** and the main support part **51** may be changed stepwise, and the user may adjust the protruding height of the gap adjusting member **40**.

(81) In addition, at least one of the plurality of main support parts **51** may support the gap adjusting member **40** by contacting the main adjusting part **41** of the gap adjusting member **40** to be described below. In other words, the height of the gap adjusting member **40** may be adjusted according to the main support part **51** to be contacted.

(82) Meanwhile, a plurality of sub support parts **52** may be formed inside the seating part **50**, that is, inside the receiving space **500**. The sub support part **52** is a portion that additionally contacts the gap adjusting member **40**, and the gap adjusting member **40** is not inclined and may maintain a more stable support state.

(83) The sub support part **52** is in contact with the sub adjusting part **42** to be described below and allows the gap adjusting member **40** to maintain a more stable support state without being inclined or deviating from a position.

(84) A plurality of sub support part **52** may be formed on each surface of the main support part **51**. For example, the sub support part **52** may be composed of three protruding ribs or protrusions on each surface of the main support part **51** and are disposed at predetermined gaps to stably support the sub adjusting part **42**. In addition, the sub support part **52** is located further outside the inclined surface **501** to be described below, and a plurality of the sub support part **52** may extend to the same position toward the center of the receiving space **500**.

(85) Accordingly, it is possible to prevent the gap adjusting member **40** from being interfered with by the sub support part **52** while the gap adjusting member **40** is detached. In addition, the sub support parts **52** may be disposed on the main support parts **51** at positions facing each other so as to face each other. In addition, the sub support part **52** may be configured to protrude from an inner surface of the main support part **51** and to protrude from each surface to a different height.

(86) In detail, a first sub support part **521** is formed on an inner surface of the third main support

part **513**, a second sub support part **522** is formed on an inner surface of the fourth main support part **514**, and a third sub support part **523** may be formed on an inner surface of the first main support part **511**, and a fourth sub support part **524** may be formed on an inner surface of the second main support part **512**.

(87) At this time, the height of the first sub support part **521** is the lowest, and the protruding height of the second sub support part **522**, the third sub support part **523**, and the fourth sub support part **524** in order becomes higher stepwise. In this case, the difference in height of the protrusions between the sub supporting parts **52** may correspond to the difference in height between the main supporting parts **51**.

(88) For example, the first sub support part **521** may have a set height and may have the lowest height among a plurality of sub support parts **52**. In addition, the second sub support part **522** is formed to be 0.5 mm higher than the first sub support part **521**, and the third sub support part **523** is formed to be 0.5 mm higher than the second sub support part **522**, and the fourth sub support part **524** may be formed to be 0.5 mm higher than the third sub support part **523**. In other words, the plurality of sub support parts **52** can be adjusted in height by one level (0.5 mm) each time they are rotated by 90° in the clockwise direction and return to the original height by rotating one turn (360°).

(89) Therefore, when the gap adjusting member **40** is further rotated stepwise by a 90° angle and then mounted at the position where the gap adjusting member **40** is initially mounted, the contact height of the sub adjusting part **42** and the sub support part **52** may be changed stepwise, and the user may adjust the protruding height of the gap adjusting member **40**.

(90) Meanwhile, a through-hole **502** penetrating the rear surface of the door **20**, that is, the rear surface of the upper cap deco **22**, may be formed in the inner center of the seating part **50**. The through-hole **502** may be formed larger than the diameter of the insertion protrusion **44** to be described below and may be formed smaller than the diameter of the elastic main support part **51**. Accordingly, when the gap adjusting member **40** is mounted, the insertion protrusion **44** may be inserted and mounted through the through-hole **502**. In addition, the elastic main support part **51** may support the circumference of the through-hole **502** in a state of passing through the through-hole **502**.

(91) In addition, an inclined surface **501** may be formed around the through-hole **502**. The inclined surface **501** may have an inclined shape such that an inner diameter becomes narrower from the front to the rear, and the through-hole **502** may be formed at a front end of the inclined surface **501**. In addition, an end portion of the sub support part **52** may be positioned at an outer end of the inclined surface **501**. Meanwhile, the inclined surface **501** may have a slope corresponding to the inclined circumferential surface of the elastic main support part **51**, and, in the process of inserting the insertion protrusion **44**, the inclined surface **501** and the circumferential surfaces of the elastic main support part **51** may be in a state of being in contact with each other and thus the insertion and mounting of the insertion protrusion **44** can be made easier.

(92) Meanwhile, the gap adjusting member **40** has a structure that can be inserted into the seating part **50** and may protrude rearward in a state of being inserted into the seating part **50**. In addition, the gap adjusting member **40** may include a contact part **45** having a planar shape, and an insertion protrusion **44** that extends from the center of the rear surface of the contact part **45** and is fixedly mounted to the seating part **50**. In addition, an adjusting part **43** that is supported by the seating part **50** and adjusts the height of the gap adjusting member **40** may be formed on the front surface of the contact part **45**.

(93) The gap adjusting member **40** may be formed of a rubber or silicone material having an elasticity as a whole, and when the door **20** is closed, the contact part **45** contacts the cabinet **10** and may mitigate the impact of the door **20**. Of course, the gap adjusting member **40** may be formed with a material having elasticity only partially, and only the elastic part **441** may be formed to have elasticity.

(94) The contact part **45** may be formed in a rectangular shape when viewed from the front and may be formed in a plate shape having a predetermined thickness. The contact part **45** may be exposed to the outside in a state where the gap adjusting member **40** is mounted on the seating part **50**. In addition, the contact part **45** may be exposed and protruded toward the rear of the seating part **50**, that is, toward the cabinet **10**.

(95) In addition, the rear surface of the contact part **45** facing the front surface of the cabinet **10** may be formed in a planar shape. In addition, a contact protrusion **451** contacting the cabinet **10** may be formed on the contact part **45**. The contact protrusion **451** may be formed along each side of the contact part **45**. In addition, the contact protrusion **451** may protrude by a predetermined height.

(96) When the door **20** is closed so that the gap adjusting member **40** contacts the front surface of the cabinet **10**, the protruding contact protrusion **451** contacts the front surface of the cabinet **10**. Accordingly, the gap adjusting member **40** may contact the front surface of the cabinet **10** in a form of line contact, and a contact area with the cabinet **10** may be reduced by the gap adjusting member **40**. In addition, even when the gap adjusting member **40** repeatedly contact the front surface of the cabinet **10** due to repetitive opening and closing of the door **20**, deformation of the front surface of the cabinet **10** or the occurrence of contact marks can be minimized.

(97) In addition, when the gap adjusting member **40** is in contact with the top cover **11** at least partially, the contact protrusion **451** also has a structure of contacting the front surface of the top cover **11** when the door **20** is closed.

(98) Meanwhile, a height indicating part **452** may be formed on the contact part **45**. The height indicating part **452** is for indicating a height set by a manipulation of the gap adjusting member **40** to a user and may be formed on four sides of the contact part **45**, respectively. For example, the height indicating part **452** may be indicated as a dot-shaped intaglio-shaped groove, and may indicate a height according to the number of grooves. The height indicating part **452** may be formed of one, two, three, and four grooves on four sides of the contact part **45**, respectively.

(99) In addition, when the gap adjusting member **40** is manipulated, the adjusted height by the gap adjusting member **40** can be easily known through the height indicating part **452** positioned at a position corresponding to the seating part guide **515** formed on the seating part **50**.

(100) For example, when the height indicating part **452** formed as a single groove is disposed at a position corresponding to the seating part guide **515**, it can be seen that the gap adjusting member **40** is set to the lowest height. In addition, when the gap adjusting member **40** is adjusted by the manipulation of the gap adjusting member **40** so that the height indicating part **452** formed of four grooves is disposed at a position corresponding to the seating part guide **515**, it can be seen that the gap adjusting member **40** is set to the highest height.

(101) Meanwhile, an adjusting member guide **415** may be formed on the contact part **45**. When the gap adjusting member **40** is mounted, the adjusting member guide **415** may assist the gap adjusting member **40** to be positioned in a correct position. The adjusting member guide **415** may be formed on one side of the circumferential surface of the contact part **45**. For example, the adjusting member guide **415** may be formed in the center of the upper surface of the four surfaces of the contact part **45** and can be disposed to be located on the same extension line as the seating part guide **515** when the gap adjusting member **40** is mounted. In addition, the surface on which the adjusting member guide **415** is formed may be a surface marked with the lowest height on the height indicating part **452**.

(102) Hereinafter, a structure for mounting and adjusting the height of the gap adjusting member **40** will be described in more detail with reference to the drawings.

(103) FIG. 7 is a perspective view illustrating the gap adjusting member viewed from one side. In addition, FIG. 8 is a perspective view illustrating the gap adjusting member viewed from the other side.

(104) As illustrated in the drawing, an adjusting part **43** may be formed on the front surface of the

contact part **45**. The adjusting part **43** supports the gap adjusting member **40** by contact with the support part **53** and can adjust the height of the gap adjusting member **40** according to the position of the support part **53** contacting the adjusting part **43**. The adjusting part **43** may include a main adjusting part **41** in contact with the main support part **51** and a sub adjusting part **42** in contact with the sub supporting part **52**.

(105) The main adjusting part **41** may be formed on the front surface of the contact part **45** and may be located more outside and more rearward than the sub adjusting part **42**. The main adjusting part **41** may be formed at a position corresponding to the protruding end portion of the main support part **51** when the gap adjusting member **40** is mounted.

(106) The main adjusting part **41** allows portions having four different heights to be disposed parallel to the four sides of the contact part **45**, and the protruding end portion of the seating part **50** disposed at a position facing each other, that is, the main support part **51** may be positioned at a position facing each other. In addition, the main adjusting part **41** may be formed along the circumference of the sub adjusting part **42** to be described below.

(107) In detail, the main adjusting part **41** may be composed of a first main adjusting part **411**, a second main adjusting part **412**, a third main adjusting part **413**, and a fourth main adjusting part **414**, along the front surface of the contact part **45**.

(108) The first main adjusting part **411** forms the same plane as the front surface of the contact part **45** and has the lowest height among the main adjusting parts **41**. In addition, the second main adjusting part **412** may extend in a direction crossing by a 90° angle from the end portion of the first main adjusting part **411** and can be formed one level higher than the first main adjusting part **411**. In addition, the third main adjusting part **413** may extend in a direction crossing by a 90° angle from the end portion of the second main adjusting part **412** and can be formed one level higher than the second main adjusting part **412**. In addition, the fourth main adjusting part **414** may extend in a direction crossing by a 90° angle from the end portion of the third main adjusting part **413**, and the end portion thereof can be formed to be connected to the first main adjusting part **411**. In addition, the fourth main adjusting part **414** may be formed one level higher than the third main adjusting part **413**.

(109) For example, the first main adjusting part **411**, the second main adjusting part **412**, the third main adjusting part **413**, and the fourth main adjusting part **414** each can be formed to be increase in height stepwise by approximately 0.5 mm. In other words, the height difference between the main adjusting parts **41** may be configured to correspond to the height difference between the main support parts **51**. Therefore, when the gap adjusting member **40** is mounted so that the adjusting member guide **415** and the seating part guide **515** are aligned, the first main support part **511** can be mounted in a state of contacting the fourth main adjusting part **414**, the second main support part **512** can be mounted in a state of contacting the third main control part **413**, the third main support part **513** can be mounted in a state of contacting the second main control part **412**, and the fourth main support part **514** can be mounted in a state of contacting the first main adjusting part **411**.

(110) Meanwhile, the sub adjusting part **42** may be further formed on the gap adjusting member **40**. The sub adjusting part **42** is in contact with the sub support part **52** so that the gap adjusting member **40** is stably supported and is located inside the main adjusting part **41** and may have a structure protruding further forward than the main adjusting part **41**.

(111) In addition, the sub adjusting part **42** is formed to be inserted into the receiving space **500** inside the seating part **50** when the gap adjusting member **40** is mounted. The sub adjusting part **42** may be formed in a shape and a size corresponding to the receiving space **500**. Therefore, when the gap adjusting member **40** is mounted on the seating part **50**, the sub adjusting part **42** is inserted into the receiving space **500**, and the outer surface of the sub adjusting part **42** can be in contact with the inner surface of the main support part **51**. Therefore, even if an impact is applied to the gap adjusting member **40** when the door **20** is closed, the gap adjusting member **40** can maintain a stable mounting state without changing or dropping the mounting position.

(112) In addition, the sub adjusting part **42** is a part supported by the sub supporting part **52** when the gap adjusting member **40** is mounted, and the main adjusting part **41** and the main support part **51** can compensate for the inability to ensure stable support.

(113) Each of the sub adjusting parts **42** may be formed along four sides of the contact part **45**, and each of the sub-adjusters **42** may have different heights. In other words, the sub adjusting part **42** can additionally support the step adjustment member **40** when the main support part **51** and the main control part **41** having different heights are rotated and mounted to each other.

(114) The sub adjusting part **42** as a whole protrudes more forward than the main control part **41** and may be formed by protruding a central part of the space formed by the main adjusting parts **41**. The sub adjusting part **42** may include a first sub adjusting part **421**, a second sub adjusting part **422**, a third sub adjusting part **423**, and a fourth sub adjusting part **424** protruding along a position corresponding to the main adjusting part, respectively.

(115) In detail, the first sub adjusting part **421** may be formed at a position corresponding to the upper surface of the contact part **45** and may protrude from a position corresponding to the third main adjusting part **413**. The first sub adjusting part **421** protrudes more forward than the main adjusting part **41** but may have the lowest height among the sub adjusting parts **42**. In addition, the protruding height of the first sub adjusting part **421** may be formed to correspond to a height difference between the second main support part **512** and the fourth sub support part **524**.

(116) The second sub adjusting part **422** may extend in a direction crossing by a 90° angle from the end portion of the first sub adjusting part **421** and may be formed one level higher than the first sub adjusting part **421**. In addition, the third sub adjusting part **423** may extend in a direction crossing by a 90° angle from the end portion of the second sub adjusting part **422** and may be formed one level higher than the second sub adjusting part **422**. In addition, the fourth sub adjusting part **424** may extend in a direction crossing by a 90° angle from the end portion of the third sub adjusting part **423** and the end portion thereof may be formed to be connected to the first sub adjusting part **421**. In addition, the fourth sub adjusting part **424** may be formed one level higher than the third sub adjusting part **423**.

(117) For example, the first sub adjusting part **421**, the second sub adjusting part **422**, the third sub adjusting part **423**, and the fourth sub adjusting part **424** each can be formed to increase in height by approximately 0.5 mm stepwise. In other words, the height difference between the sub adjusting parts **42** may be configured to correspond to the height difference between the sub support parts **52**. When the gap adjusting member **40** is mounted so that the adjusting member guide **415** and the seating part guide **515** are aligned, the first sub adjusting part **421** can be mounted in a state of being in contact with the fourth sub support part **524**, the second sub adjusting part **422** can be mounted in a state of being in contact with the third sub supporting part **523**, the third sub adjusting part **423** can be mounted in a state of being in contact with the second sub supporting part **522**, and the fourth sub adjusting part **424** can be mounted in a state of being in contact with the first sub adjusting part **521**.

(118) An insertion protrusion **44** protruding forward may be formed in the center of the sub adjusting part **42**. The insertion protrusion **44** is for fixing and mounting the gap adjusting member **40** to the rear surface of the door **20** and can be formed to extend forward from the central portion of the space formed by the sub adjusting part **42**. The extension length of the sub adjusting part **42** is such that the elastic part **441** formed in the insertion protrusion **44** can pass through the through-hole **502** and protrude by a length that can be inserted into the door **20**.

(119) The elastic part **441** (restrain part) may be formed to protrude outward from one side of the insertion protrusion **44**, and in a state where the gap adjusting member **40** is mounted on the door **20**, the gap adjusting member **40** can be prevented from being separated by being locked by the circumference of the through-hole **502**. In addition, by the elastic force provided by the elastic part **441**, the gap adjusting member **40** may be returned to the inserted state in the seating part **50** after manipulation.

(120) In detail, the elastic part **441** may have a structure that protrudes outward from one side of the insertion protrusion **44** along the circumference of the insertion protrusion **44** and extends rearward. At this time, the connection position of the elastic part **441** and the insertion protrusion **44** may be determined as a position in which the elastically deformed state can be maintained while the elastic part **441** does not fall out through the through-hole **502** when the sub adjusting part **42** pulls the gap adjusting member **40** so as to deviate from the seating part **50**.

(121) In addition, the elastic part **441** may be formed to be further away from the outer surface of the insertion protrusion **44** as it extends rearward. In other words, the circumferential surface of the elastic part **441** may be formed to have an inclination and may be formed to have an inclination corresponding to the inclined surface **501**.

(122) In addition, the elastic part **441** may be formed to a thickness that is easily elastically deformed. Therefore, when the gap adjusting member **40** is mounted on the seating part **50**, the elastic part **441** is elastically deformed and moved along the inclined surface **501**, and then can pass through the through-hole **502**. In addition, in a state where the gap adjusting member **40** is mounted, the elastic part **441** is elastically deformed according to the manipulation of the gap adjusting member **40**, so that it may be possible to move the step adjustment member **40** in the front and rear direction and return to the initial position by elasticity.

(123) Hereinafter, a state where the gap adjusting member **40** is mounted will be described in more detail with reference to the drawings.

(124) FIG. **9** is a cut-away perspective view illustrating the gap adjusting member in a seated state.

(125) As illustrated in the drawing, the gap adjusting member **40** may be mounted on the rear surface of the door **20** in a state of being seated on the seating part **50**, and protrudes from the rear surface of the door **20** to the rear and thus when the door **20** is closed, the door contacts the cabinet **10** so that the rotation of the door **20** is stopped.

(126) In order to mount the gap adjusting member **40** to the seating part **50**, first, the insertion protrusion **44** is inserted into the through-hole **502** while the adjusting member guide **415** and the seating part guide **515** are aligned. Further, the insertion protrusion **44** is inserted until the elastic part **441** passes through the through-hole **502**, and when the elastic part **441** passes through the through-hole **502**, the insertion of the gap adjusting member **40** may be completed.

(127) In addition, when the gap adjusting member **40** is mounted on the seating part **50**, the sub adjusting part **42** of the gap adjusting member **40** can be inserted in the receiving space **500** inside the seating part **50**. In a state where the adjusting member guide **415** and the seating part guide **515** are aligned to be positioned on the same line, each part of the main adjusting part **41** and the sub adjusting part **42** of the gap adjusting member **40** may be mounted to be in contact with the main support part **51** and the sub support part **52** of the seating part **50**, respectively.

(128) Accordingly, the gap adjusting member **40** is mounted on the seating part **50** at an accurate position and can stably support the cabinet **10** at a set height by maintaining the mounted state. In addition, in the state where the gap adjusting member **40** is mounted, the extended rear end of the elastic part **441** is supported on the inside surface of the door **20**, that is, the inside surface of the upper cap deco **22**.

(129) Meanwhile, in the state as illustrated in FIG. **9**, if the user wants to adjust the degree of protrusion of the front surface of the door **20** while the gap adjusting member **40** is manipulated to have the third highest height, the height of the gap adjusting member **40** is manipulated.

(130) In order to adjust the height of the gap adjusting member **40**, the user can pull the contact part **45** of the gap adjusting member **40** rearward and rotates the contact part **45** by a set angle and then can mount on the seating part **50**. In this case, the gap adjusting member **40** may be rotated about the insertion protrusion **44** as a shaft. Through such a simple operation, the height of the gap adjusting member **40** can be adjusted from the rear surface of the door **20**.

(131) Hereinafter, the manipulation state of the gap adjusting member **40** will be described with reference to the drawings.

(132) In addition, FIG. 10 is a perspective view illustrating a state where the gap adjusting member is pulled out for height adjusting. In addition, FIG. 11 is a cut-away perspective view of FIG. 10. (133) As illustrated in the drawings, in order to adjust the height of the gap adjusting member 40, the gap adjusting member 40 is pulled rearward. The gap adjusting member 40 may be moved rearward until the sub adjusting part 42 completely deviates from the opened rear surface of the seating part 50.

(134) When the gap adjusting member 40 is pulled rearward, the insertion protrusion 44 is also moved rearward, and the elastic part 441 supported by the circumferential surface of the through-hole 502 is elastically deformed. The elastic part 441 may be elastically deformed while being pressed relatively forward as the gap adjusting member 40 is moved backward.

(135) Accordingly, the elastic part 441 is elastically deformed outward by the pressure of the rear surface of the door 20 and maintains a contact state with the rear surface of the door 20. In addition, when the sub adjusting part 42 is moved rearward until it is located deviating from the receiving space 500, the pressure of the elastic part 441 is maximized, and the elastic part 441 is also can be elastically deformed in an extended state.

(136) The user pulls the sub adjusting part 42 backward and pulls the sub adjusting part 42 out to the opened rear surface of the receiving space 500 and then rotates the contact part 400 in a desired direction by a set angle. Whenever the contact part 400 is rotated at a set angle (in units of 90°), it is possible to adjust the height by one level and may be set to a desired height.

(137) In this case, the user can determine the set height of the gap adjusting member 40 by looking at the height indicating part 452. In other words, the user may rotate the gap adjusting member 40 so that the height indicating part 452 having a desired height is positioned on the same extension line as the seating part guide 515.

(138) When the contact part 400 is placed in a state where the gap adjusting member 40 is rotated to a height desired by the user, the sub adjusting part 42 can be inserted into the inside of the receiving space 500 by the restoring force of the extended elastic part 441. In addition, the main adjusting part 41 and the sub adjusting part 42 of the gap adjusting member 40 are supported by the main support part 51 and the sub support part 52 at a set position, respectively, and the gap adjusting member 40 can be the set height.

(139) Hereinafter, the support state of the gap adjusting member 40 and the seating part 50 at each height at which the gap adjusting member 40 is manipulated will be described in more detail with reference to the drawings.

(140) FIG. 12 is a cut-away perspective view illustrating the first level height adjusting state of the gap adjusting member in one step.

(141) As illustrated in the drawing, when the gap adjusting member 40 is first assembled to the door, it may be assembled in the first level height state as illustrated in FIG. 12. At this time, when the gap adjusting member 40 is mounted in a case where the adjusting member guide 415 and the seating part guide 515 are disposed on the same line, it can be checked that the gap adjusting member 40 is mounted at the correct position.

(142) In addition, the height indicating part 452, which is a standard for determining the height of the gap adjusting member 40, may be determined based on that disposed on an extension line of the seating part guide 515. In other words, the height of the gap adjusting member 40 is determined by reading the uppermost height indicating part 452. In FIG. 12, it may be recognized that the height indicating part 452 including one groove is positioned at the uppermost position and has the lowest height. For example, as illustrated in FIG. 12, at the first level of height D1, a height of approximately 10.5 mm may be obtained from the rear surface of the door 20 to the rear end of the gap adjusting member 40. In other words, the distance between the rear surface of the door 20 and the cabinet 10 may be adjusted to 10.5 mm.

(143) In a state where the gap adjusting member 40 is mounted, the sub adjusting part 42 is inserted into the receiving space 500 of the seating part 50 to maintain a stable mounting state. At this time,

the sub adjusting part **42** having a rectangular cross-section is disposed inside the receiving space **500** having the same cross-sectional structure to maintain a stable mounting state without causing flow or deviating from the mounting position.

(144) In addition, the control part **43** and the support part **53** as a whole have a stable support structure in contact with each other. In detail, the third main control part **413** and the first sub adjusting part **421** positioned at the upper side of the adjusting part **43** are in a state of being supported by contacting the second main support part **512** and the fourth sub support part **524** located at the upper side of the support part **53**, respectively.

(145) The first main control part **411** and the third sub adjusting part **423** positioned at the lower side of the control part **43** are in a state of being supported by contacting the fourth main support part **514** located at the lower side of the support part **53** and the second sub supports **522**, respectively.

(146) Further, although not illustrated in detail, the adjusting parts **43** and the support parts **53** disposed on both left and right sides may also be supported by each other at positions facing each other. In other words, the fourth main adjusting part **414** and the second sub adjusting part **422** positioned on the left side may be supported by contacting the first main support part **511** and the third sub support part **523**, respectively. In addition, the second main adjusting part **412** and the fourth sub adjusting part **424** positioned on the right side may be supported by contacting the third main support part **513** and the first sub support part **521**, respectively.

(147) In this way, in a state where the gap adjusting member **40** is adjusted to a first-level height **D1**, all four surfaces of the upper, lower, left, and right sides of the adjusting part **43** are in contact with all four surfaces of the support part **53** and thus can maintain a stable support state.

(148) In this way, the control part **43** and the support part **53** have a structure in which the entire surfaces facing each other are supported. Therefore, even if an impact or a load is applied to the contact part **45** to an eccentric position, the gap adjusting member **40** can be maintained without deviating from the mounting position.

(149) In addition, although not illustrated, the adjusting parts **43** and the support parts disposed on both left and right sides are also supported at positions facing each other, so that the entire four surfaces can maintain a stable support state.

(150) Meanwhile, in such a state, when the height of the front surface of the door **20** is low and a step occurs with the front surface of another door **20** or home appliance, furniture or wall **O** adjacent thereto, the gap adjusting member **40** is manipulated to increase the height thereof.

(151) To this end, the user grabs the exposed contact part **45** as illustrated in FIG. **11** and pulls the contact part backward, so that the adjusting part **43** is deviated from the inside of the seating space, and then rotates and then the user releases the adjusting part, and by the elastic force of the elastic part **441**, the gap adjusting member **40** can be mounted back to the seating part **50**. At this time, the height of the gap adjusting member **40** to be adjusted can be checked through the height indicating part **452**, and the cross-sectional structure of the adjusting part **43** and the seating part is made of a polygonal structure, so that the combination of the adjusting part **43** and the seating part at the correct position can be induced.

(152) FIG. **13** is a cut-away perspective view illustrating a two level height adjusting state of the gap adjusting member.

(153) As illustrated in the drawing, the user can manipulate the gap adjusting member **40** to have a second step height by performing a manipulation of rotating the gap adjusting member **40** clockwise by 90° based on FIG. **12**. For example, in the second level height **D2**, a height of about 11 mm may be obtained from the rear surface of the door **20** to the rear end of the gap adjusting member **40**. In other words, the gap between the rear surface of the door **20** and the front surface of the cabinet **10** may be adjusted to 11 mm, and it can be seen that the height increases by one level (5 mm) compared to the first step height.

(154) Looking at the states of the adjusting part **43** and the support part **53** at this time, the fourth

main adjusting part **414** and the second sub adjusting part **422** among the adjusting parts **43** are located on the upper side by the rotation of the gap adjusting member **40**. In addition, the fourth main adjusting part **414** and the second sub adjusting part **422** are in a state of being supported by contacting the second main support part **512** and the fourth sub support part **524**, respectively, positioned on the upper side of the support part **53**.

(155) In addition, the second main adjusting part **412** and the fourth sub adjusting part **424** among the adjusting parts **43** are positioned at the lower side by the rotation of the gap adjusting member **40**. In addition, the second main adjusting part **412** and the fourth sub adjusting part **424** are in a state of contacting the fourth main support part **514** and the second sub support part **522** located at the lower side of the support part **53**, respectively.

(156) In addition, although not illustrated in detail, the third main adjusting part **413** and the first sub adjusting part **421** on the right side of the adjusting parts **43** and the support parts **53** disposed on both left and right sides may be disposed to face each other with the third main support part **513** and the first sub support **521**. At this time, the third main adjusting part **413** and the third main supporting part **513** are in contact with each other, but the first sub adjusting part **421** and the first sub supporting part **521** have a low protruding height and thus can be in a state of being spaced apart from each other.

(157) In addition, the first main adjusting part **411** and the third sub adjusting part **423** on the left side of the adjusting parts **43** and the support parts **53** disposed on both left and right sides may be disposed to face each other with the first main support part **511** and the third sub support part **523**. At this time, the first main control part **411** and the first main support part **511** have a low protruding height and are spaced apart from each other, but the third sub adjusting part **423** and the third sub support part **523** may be in a state of being in contact with each other.

(158) In this way, the adjusting part **43** and the support part **53** have a structure that supports each other at the upper and lower positions, which are positions facing each other on the upper, lower, left, and right four surfaces. Therefore, even if an impact or a load is applied to the contact part **45** to an eccentric position, the gap adjusting member **40** can be maintained without deviating from the mounting position.

(159) In addition, compared with the first level height of FIG. **12**, the position of the adjusting part **43** supported by the seating part **50** by 90° rotation of the gap adjusting member **40** can be changed to the adjusting parts **43** that are one level higher. Accordingly, the distance D2 (11 mm) between the rear surface of the door **20** and the cabinet **10** increases, and the front surface of the door **20** can be further protruded.

(160) At this time, when the gap adjusting member **40** is rotated, the user can check the height of two levels by placing the height indicating part **452** consisting of two grooves on the upper side, and by inserting the adjusting part **43** having a polygonal shape into the receiving space **500**, the correct height to be desired can be adjusted.

(161) FIG. **14** is a cut-away perspective view illustrating a third level height adjusting state of the gap adjusting member.

(162) As illustrated in the drawing, the user can manipulate the gap adjusting member **40** to have a third level height by performing a manipulation of rotating the gap adjusting member **40** by 180° in the clockwise direction with reference to FIG. **12**. For example, in the third level height D3, a height of about 11.5 mm may be obtained from the rear surface of the door **20** to the rear end of the gap adjusting member **40**. In other words, the gap between the rear surface of the door **20** and the cabinet may be adjusted to 11.5 mm, and it can be seen that the height increases by two levels (10 mm) compared to the height of the first level.

(163) Looking at the states of the adjusting part **43** and the support part **53** at this time, the first main adjusting part **411** and the third sub adjusting part **423** among the adjusting parts **43** by the rotation of the gap adjusting member **40** are located on the upper side. In addition, the first main adjusting part **414** and the third sub adjusting part **422** are positioned at a position which faces each

other with the second main support part **512** and the fourth sub support part **524** positioned on the upper side of the support part **53**.

(164) In this case, the first main adjusting part **411** having a relatively low protrusion height is spaced apart from the second main support part **512**, and the third sub adjusting part **423** having a relatively high protruding height may be in a state of being supported in contact with the forth sub support part **524**.

(165) In addition, the third main adjusting part **413** and the first sub adjusting part **421** among the adjusting parts **43** are positioned at the lower side by the rotation of the gap adjusting member **40**. Among these, the first sub adjusting part **421** and the second sub support part **522** having a relatively low protrusion height are spaced apart from each other, and the third main control part **413** can be in a supported state by being in contact with the fourth main support part **514**.

(166) In addition, although not illustrated in detail, the fourth main adjusting part **414** and the second sub adjusting part **422** of the right side among the adjusting parts **43** and the support part **53** placed on both sides of the left and right are disposed to face each other with the third main support part **513** and the first sub support part **521**. In this case, the fourth main adjusting part **414** and the third main support part **513** are in contact with each other, but the second sub-adjusting part **422** and the first sub-supporting part **521** may be a state of having a low protruding height so that they will be spaced apart from each other.

(167) In addition, the second main adjusting part **412** and the fourth sub adjusting part **424** on the left of the adjusting parts **43** and the support parts **53** disposed on both sides of the left and right may be disposed to face each other with the first main support part **511** and the third sub support part **523**. At this time, the second main adjusting part **412** and the first main support part **511** are spaced apart from each other due to their low protruding height, but the fourth sub adjusting part **424** and the third sub support part **523** are in a state of being in contact with each other.

(168) In this way, the adjusting part **43** and the support part **53** have a structure that supports each other on the upper, lower, left, and right four surfaces. Therefore, even if an impact or a load is applied to the contact part **45** to an eccentric position, the gap adjusting member **40** can be maintained without leaving the mounting position.

(169) In addition, compared with the first level height of FIG. **12**, the position of the adjusting part **43** supported by the seating part **50** by 180° rotation of the gap adjusting member **40** can be changed to the third sub adjusting part **423** and the third main adjusting part **413**, which is two levels higher. Accordingly, the distance D3 (11.5 mm) between the rear surface of the door **20** and the cabinet **10** increases and the front surface of the door **20** can be further protruded.

(170) At this time, when the gap adjusting member **40** is rotated, the user can check the third level height by placing the height indicating part **452** consisting of three grooves on the upper side, and the polygonal adjusting part **43** and it is possible to adjust to the desired correct height by inserting the adjusting part **43** inserted into the receiving space **500**.

(171) FIG. **15** is a cut-away perspective view illustrating a fourth level height adjusting state of the gap adjusting member.

(172) As illustrated in the drawing, the user can manipulate the gap adjusting member **40** to have the fourth level height by performing a manipulation of rotating the gap adjusting member **40** by 270° in the clockwise direction based on FIG. **12**. For example, at the fourth level height D3, a height of about 12 mm may be obtained from the rear surface of the door **20** to the rear end of the gap adjusting member **40**. In other words, the gap between the rear surface of the door **20** and the cabinet may be adjusted to 12 mm, and it can be seen that the height increases by two levels (10 mm) compared to the first step height.

(173) Looking at the states of the adjusting part **43** and the support part **53** at this time, the second main adjusting part **412** and the fourth sub adjusting part **424** among the adjusting parts **43** are positioned on the upper side by the rotation of the gap adjusting member **40**, and the second main control part **412** and the fourth sub adjusting part **424** are in a state where the second main support

part **512** and the fourth sub support part **524** faces each other with the second main support part **512** and the fourth sub support part **524**, respectively.

(174) In this case, the second main adjusting part **412** and the second main support part **512** are spaced apart from each other due to a relatively low protrusion height, and the fourth sub adjusting part **424** having a relatively high protrusion height may be in a state of being supported in contact with the sub supporting part **524**.

(175) In addition, by the rotation of the gap adjusting member **40**, the fourth main adjusting part **414**, and the second sub adjusting part **422** of the adjusting part **43** are located at the lower side, and the fourth main adjusting part **414** and the second sub adjusting part **422** are in a state of facing each other with the fourth main support part **514** and the second sub support part **522**, respectively.

(176) In this case, the fourth main adjusting part **414** and the fourth main supporting part may be supported with each other, and the second sub adjusting part **422** having a relatively low protruding height may be spaced apart from the second sub support part **522**.

(177) In addition, although not illustrated in detail, the first main adjusting part **411** and the third sub adjusting part **423** on the right side of the adjusting parts **43** and support parts **53** disposed on both left and right sides may be disposed to face each other with the main support part **513** and the first sub support part **521**, respectively. In this case, both of the first main adjusting part **411** and the third sub adjusting part **423** may be in a state spaced apart from the third main support part **513** and the first sub support part **521**.

(178) In addition, the third main adjusting part **413** and the first sub adjusting part **421** on the left of the adjusting parts **43** and the support parts **53** disposed on both left and right sides may be disposed to face each other with the first main support part **511** and the third sub support part **523**. In this case, both of the third main adjusting part **413** and the first sub adjusting part **421** may be in a state of being spaced apart from the first main support part **511** and the third sub support part **523** from each other.

(179) In this way, the adjusting part **43** and the support part **53** have a structure that supports each other on the upper and lower surfaces. Although the adjusting part **43** and the support part **53** come into contact with each other only on two surfaces unlike in the above levels 1 to 3, the adjusting part **43** and the support part **53** have a structure that is supported in the vertical direction facing each other since the fourth sub adjusting part **424** is supported by the fourth sub support part **524** from the upper side, and the fourth main adjusting part **414** is supported by the fourth main support part **514** from the lower side. Therefore, even if an impact or a load is applied to the contact part **45** to an eccentric position, the gap adjusting member **40** can be maintained without deviating from the mounting position.

(180) In addition, compared with the first level height of FIG. **12**, the position of the adjusting part **43** supported by the seating part **50** by 270° rotation of the gap adjusting member **40** can be changed to the fourth sub adjusting part **424** and the fourth main control part **414** which are three levels higher. Accordingly, the distance D4 (12 mm) between the rear surface of the door **20** and the cabinet **10** increases, and the front surface of the door **20** can be further protruded.

(181) At this time, when the gap adjusting member **40** is rotated, the user can check the fourth level height by placing the height indicating part **452** consisting of four grooves on the upper side, and it is possible to adjust the height to a desired exact height by inserting the polygonal adjusting part **43** into the receiving space **500**.

(182) Meanwhile, by using the gap adjusting member **40** whose height is adjusted in the same manner as described above the height of the front surface of the door **20** in the front and rear direction can be adjusted to have the same height in the front surface of adjacent door **31**, **32**, **33**, and **34**, furniture, or wall O.

(183) FIG. **16** is a view schematically illustrating a state of adjusting a step between the door and an adjacent configuration.

(184) As illustrated in the drawing, when the door **20** is closed while the refrigerator **1** is installed,

the front surface of the door **34** may protrude by the front surface of the door **34** of the refrigerator **1'** adjacent to the door **34** or a difference in height H from the front surface of the door **34** of another refrigerator **1'**. Further, in such a state, a step may be generated between the door and the neighboring door **34**, the furniture, or wall, thereby deteriorating the outer appearance.

(185) At this time, the user opens the door **20** to adjust the protruding height of the door **20** and then manipulates the gap adjusting member **40** mounted on the rear surface **222** of the door **20** and thus the protruding height of the gap adjusting member **40** can be lowered.

(186) The user pulls the gap adjusting member **40** to pull the adjusting part **43** out of the receiving space. Even in such a state, the gap adjusting member **40** remains a state of being mounted on the door **20**, but the elastic part **441** may be elastically deformed. When the user confirms that the height is rotated to a desired height through the height indicating part **415** and then releases the gap adjusting member **40**, the adjusting part **43** can be inserted into the inside of the receiving space **50** by the elastic force of the elastic part **441**.

(187) Through such adjusting of the protruding height of the gap adjusting member **40**, the gap D between the rear surface of the door **20** and the front surface of the cabinet **10** can be adjusted to become narrower. Accordingly, the front surface of the door **20** is lowered by the adjusted height and may be finally located on the same plane as the front surface of the neighboring door **34** or the furniture or wall O.

(188) Meanwhile, the gap adjusting member according to the embodiment of the present disclosure may be various other embodiments in addition to the above-described embodiment. The gap adjusting member according to another embodiment of the present disclosure may have a structure in which a contact part is formed in a circular shape, and a shield portion is further formed in the contact part.

(189) In addition, the gap adjusting member according to another embodiment of the present disclosure differs only in the shape of the contact part and the shield part, but all other configurations are the same, and the same configurations as in the above-described embodiment use the same reference numerals, and detailed descriptions thereof can be omitted or the illustration thereof can be omitted.

(190) FIG. **17** is a perspective view illustrating a refrigerator door according to another embodiment of the present disclosure as viewed from the rear. In addition, FIG. **18** is an exploded perspective view illustrating the coupling relationship between the gap adjusting member and the seating part. In addition, FIG. **19** is a cut-away perspective view taken along line XIX-XIX' of FIG. **17**. In addition, FIG. **20** is a cut-away perspective view illustrating a state where the gap adjusting member is pulled out for height adjustment in FIG. **15**.

(191) As illustrated in the drawings, the outer appearance of the refrigerator door **20** according to another embodiment of the present disclosure, like the door of the above-described embodiment, may be formed by a front plate **21** forming a front surface and a door liner **25** forming a rear surface, and an upper cap deco **22** and a lower cap deco **23** forming the upper and lower surfaces. In addition, the side frames **24** may be further provided on both left and right side surfaces of the door **20**.

(192) A part of the rear surface of the door **20** may be formed by the rear surface of the upper cap deco **22**. A rear surface of the upper cap deco **22** may be disposed above the gasket **26** mounted on the door liner **25**.

(193) A gap adjusting member **60** according to another embodiment of the present disclosure may be provided on the rear surface of the door **20**, that is, the rear surface of the upper cap deco **22**. The gap adjusting member **60** may protrude from the rear surface of the door **20**. In addition, the gap adjusting member **60** may be provided on a side far from the hinge mounting part **221** of the both left and right sides of the upper cap deco **22**.

(194) In order to mount the gap adjusting member **60**, a seating part **50** may be formed on the rear surface of the upper cap deco **22**. The structure and shape of the seating part **50** may be completely

the same as in the above-described embodiment.

(195) The seating part **50** may have a polygonal cross-sectional shape and may be formed to have an opened rear surface. In addition, a receiving space **500** may be formed inside the seating part **50**, and an inclined surface **501** and a through-hole **502** may be formed in the center, respectively.

(196) In addition, a support part **53** may be formed on the seating part **50**. The support part **53** may include a main support part **51** forming an outer circumference shape of the seating part **50** and a sub support part **52** inside the main support part **51**.

(197) The main support part **51** includes a first main support part **511**, a second main support part **512**, a third main support part **513**, and a fourth main support part **514**, and may be formed in a rectangular shape. In addition, the first main support part **511** has the lowest height, and the height of the second main support part **512**, the third main support part **513**, and the fourth main support part **514** can be formed to increase stepwise in order.

(198) The sub support part **52** protrudes from an inner surface of the main support part **51** and may be disposed inside the receiving space **500**. The sub support part **52** may protrude vertically from each surface of the main support part **51** and may be formed of a plurality of protrusions.

(199) The sub support part **52** may include a first sub support part **521** protruding from an inner surface of the third main support part **513**, a second sub support part **522** protruding from an inner surface of the fourth main support part **514**, and a third sub support part **523** protruding from the inner surface of the first main support part **511**, and a fourth sub support part **524** protruding from the inner surface of the second main support part **512**. In addition, the first sub support part **521** has the lowest height, and the height of the second sub support part **522**, the third sub support part **523**, and the fourth sub support part **524** can increase stepwise.

(200) In addition, a seating part guide **515** may be formed on an outer surface of the second sub support part **522**.

(201) Meanwhile, the gap adjusting member **60** may include a contact part **65** forming a rear surface in contact with the cabinet **10**, a shielding portion **66** extending forward along the circumference of the contact part **65**, and an adjusting part **43** and an insertion protrusion **44** formed on the front surface of the contact part **65**. In addition, an elastic part **441** may be formed in the insertion protrusion **44**.

(202) The gap adjusting member **60** may be formed of at least a part including the elastic part **441** made of a rubber or silicone material and may be elastically deformed when the gap adjusting member **60** is pulled to provide an elastic force. In addition, at least a part of the gap adjusting member **60** may be formed of an elastic material to mitigate an impact generated upon contact with the cabinet **10**.

(203) The contact part **65** forms a rear surface in contact with the cabinet **10** and may be formed in a planar shape. In addition, a height indicating part **652** may be formed on the contact part **65**. The height indicating part **652** may protrude in a protrusion shape, and the height may be indicated by the number of protrusions.

(204) In this case, the height indicating part **652** may be arranged to rotate at 90° angle intervals based on the center of the contact part **65**. In addition, the height indicating part **652** may be positioned at a position corresponding to the position of the adjusting part **43** to indicate the height of the corresponding adjusting part **43** to the user.

(205) In addition, since the height indicating part **652** protrudes in a protruding shape, the height indicating part **652** may contact the front surface of the cabinet **10** when the door is closed.

Accordingly, the contact area between the gap adjusting member **60** and the cabinet **10** is minimized, and deformation and damage of the front surface of the cabinet **10** are prevented.

(206) In addition, a shielding portion **66** extending forward along the periphery of the contact part **65** may be formed. The shielding part **66** may extend from the contact part **65** to the rear surface of the door **20**. Therefore, it is possible to shield the seating part **50** from being exposed to the outside in a state where the gap adjusting member **60** is mounted.

(207) The outer surface of the shielding part **66** may be formed in a circular shape, and when the gap adjusting member **60** is manipulated to the lowest height, the extension length of the shielding part **66** can be formed to a length that does not interfere to the rear surface of the door **20**.

(208) In addition, the inner surface of the shielding portion **66** may be formed in a shape corresponding to the shape of the seating part **50** having a polygonal shape. Accordingly, in a state where the gap adjusting member **60** is mounted, the inner surface of the shielding part **66** may contact the inner surface of the seating part **50**. In other words, the shielding part **66** may be opened rearward, and the seating part **50** is inserted into the shielding part **66** so that the seating part **50** can contact the inside of the shielding part **66**.

(209) Meanwhile, an adjusting part **43** and the insertion protrusion **44** may be formed inside the shielding part **66**, that is, on the front surface of the contact part **65**. The shape and structure of the adjusting part **43** and the insertion protrusion **44** may be the same as those of the above-described embodiment, and reference numerals not illustrated may refer to the above-described embodiments.

(210) The adjusting part **43** may include a main adjusting part **41** protruding from a position corresponding to the main support part **51** and a sub adjusting part **42** protruding from a position corresponding to the sub support part **52**.

(211) The main adjusting part **41** is disposed along the circumference of the sub adjusting part **42**, in order of height, the first main adjusting part **411**, the second main adjusting part **412**, and the third main adjusting part **413**, and a fourth main adjusting part **414**. In addition, the sub adjusting part **42** is located inside the main adjusting part **41** and may include the first sub adjusting part **421** inside the third main adjusting part **413**, and the fourth main adjusting part **414** inside a second sub adjusting part **422**, a third sub adjusting part **423** inside the first main control part **411**, and a fourth sub adjusting part **424** inside the second main control part **412**.

(212) The structure of the adjusting part **43** and the structure of the seating part **50** in contact with the adjusting part **43** may be the same as those of the above-described embodiment, and reference numerals not illustrated can refer to the above-described embodiments. Further, the position of the adjusting part **43** in contact with the seating part **50** is changed by the rotation of the adjusting part **43** so that the height of the gap adjusting member **60** can be adjusted.

(213) Meanwhile, the insertion protrusion **44** may extend forward from the center of the sub adjusting part **42** and penetrate the through-hole **502** to be located inside the door **20**. In addition, an elastic part **441** may be formed around the insertion protrusion **44**. The elastic part **441** may be inserted along the inclined surface **501** to elastically support the rear surface of the door **20** while being positioned in front of the through-hole **502**.

(214) The gap adjusting member **60** having such a structure may be able to adjust the height by a simple manipulation without separate separation and disassembly manipulation.

(215) When the door **20** is opened, the gap adjusting member **60** is pulled rearward, as illustrated in FIG. **20**, the adjusting part **43** pulls out to the opened rear surface of the seating part **50**. In this case, the elastic part **441** may be elastically deformed by being supported by the rear surface of the door **20**.

(216) In a state where the adjusting part **43** is moved outside the receiving space, the user can change the position of the adjusting part **43** by rotating the contact part **65**. In other words, by changing the disposition of the adjusting part **43** seated on the seating part **50** by the rotation of the contact part **65**, the support part **53** having a different height can be supported by the seating part **50**.

(217) When the contact part **65** is rotated so that the support part **53** can be disposed at a desired position, the gap adjusting member **60** may be rotated about the insertion protrusion **44**. In addition, the selected height can be checked through the height indicating part **652**.

(218) When the gap adjusting member **60** is released after rotating the adjusting part **43** to a desired height, the adjusting part **43** may be inserted into the receiving space of the seating part **50** by an elastic force. In addition, the adjusting part **43** is supported by the support part **53** and at the same

time can be inserted into the receiving space **500** so that the gap adjusting member **60** can maintain a stable mounting state.

(219) Meanwhile, a refrigerator according to an embodiment of the present disclosure may have various other embodiments in addition to the above-described embodiments. A refrigerator according to another embodiment of the present disclosure is characterized in that a gap adjusting member having a circular shape on a rear surface in contact with the cabinet and a seating part on which the gap adjusting member is seated have the same outer diameter. In addition, the refrigerator according to another embodiment of the present disclosure has only the shape of the contact part and the outer shape of the seating part among the gap adjusting members, all other configurations are the same, the same configurations as those of the above-described embodiment use the same reference numerals, and the detailed description may be omitted or the illustration thereof may be omitted.

(220) FIG. **21** is a partial perspective view illustrating a state where a gap adjusting member is mounted according to another embodiment of the present disclosure. In addition, FIG. **22** is an exploded perspective view illustrating the gap adjusting member of the present disclosure separated from the seating part. In addition, FIG. **23** is a cut-away perspective view taken along the line XXIII-XXIII' of FIG. **21**.

(221) As illustrated in the drawing, a gap adjusting member **70** contacting the cabinet **10** may be provided on the rear surface of the door **20** of the refrigerator according to another embodiment of the present disclosure when the door **20** is closed.

(222) The gap adjusting member **70** may be mounted on the seating part **50** formed on the rear surface of the upper cap deco **22**. In addition, as in the above-described embodiment, the gap adjusting member **70** may be rotated while being pulled out to the rear, and then seated on the seating part **50** again, thereby adjusting the height.

(223) In detail, the gap adjusting member **70** may include a contact part **75** forming a rear surface in contact with the cabinet **10**, and an adjusting part **43** and an insertion protrusion **44** formed on the front surface of the contact part **75**. In addition, an elastic part **441** may be formed in the insertion protrusion **44**.

(224) The gap adjusting member **70** may be formed of at least a part including the elastic part **441** of a rubber or silicone material to be elastically deformed when the gap adjusting member **70** is pulled to provide an elastic force. In addition, at least a part of the gap adjusting member **70** may be formed of an elastic material to mitigate an impact generated when contacting the cabinet **10**.

(225) The contact part **75** forms a rear surface in contact with the cabinet **10** and may be formed in a planar shape. The contact part **75** may be formed in a circular shape corresponding to the size of the seating part **50**, and may be formed to have a predetermined thickness to mitigate an impact when the door **20** is closed.

(226) In addition, a height indicating part **752** may be formed on the rear surface of the contact part **75**. The height indicating part **752** may be recessed, and the height may be displayed by the number of recessed spots. The height indicating part **752** may be arranged to be rotated at 90° angle intervals based on the center of the contact part **75**. In addition, the height indicating part **752** may be positioned at a position corresponding to the position of the adjusting part **43** to display the height of the corresponding adjusting part **43** to the user.

(227) Meanwhile, an adjusting part **43** and the insertion protrusion **44** may be formed on the front surface of the contact part **65**. The shape and structure of the adjusting part **43** and the insertion protrusion **44** may be the same as those of the above-described embodiment, and reference numerals not illustrated may refer to the above-described embodiments.

(228) The adjusting part **43** may include a main adjusting part **41** protruding from a position corresponding to the main support part **51** of the seating part **50** and a sub adjusting part **42** protruding from the position corresponding to the sub supporting part **52** of the seating part **50**.

(229) The main adjusting part **41** is disposed along the circumference of the sub adjusting part **42**,

and may include, in order of height, the first main adjusting part **411**, the second main adjusting part **412**, and the third main adjusting part **413**, and a fourth main adjusting part **414**. In addition, the sub adjusting part **42** is located inside the main adjusting part **41** and may include the first sub adjusting part **421** inside the third main adjusting part **413**, and a second sub adjusting part **422** inside the fourth main adjusting part **414**, a third sub adjusting part **423** inside the first main control part **411**, and a fourth sub adjusting part **424** inside the second main control part **412**.

(230) The structure of the adjusting part **43** and the structure of the seating part **50** in contact with the adjusting part **43** may be the same as those of the above-described embodiment, and reference numerals not illustrated can refer to the above-described embodiments. In addition, the position of the adjusting part **43** in contact with the seating part **50** is changed by the rotation of the adjusting part **43** so that the height of the gap adjusting member **70** can be adjusted.

(231) In addition, the insertion protrusion **44** may extend forward from the center of the sub adjusting part **42** and is positioned inside the door **20** through the through-hole **502**. In addition, an elastic part **441** may be formed around the insertion protrusion **44**. The elastic part **441** may be inserted along the inclined surface **501** to elastically support the rear surface of the door **20** while being positioned in front of the through-hole **502**.

(232) Meanwhile, in order to mount the gap adjusting member **70**, a seating part **50** may be formed on the rear surface of the upper cap deco **22**. The rest of the structure and shape of the seating part **50** except for the outer seating part edge **54** may be completely the same as in the above-described embodiment.

(233) The seating part **50** may include a seating part edge **54** having an outer diameter equal to the outer diameter of the contact part **75**. The seating part edge **54** may form an outer circumferential surface of the seating part **50**. The seating part edge **54** may be formed to have a circular cross section of a size and shape corresponding to the contact part **75** and can protrude rearward by the predetermined height so that the receiving space **500** and the support part **53** may be formed therein.

(234) The receiving space **500** having a polygonal shape corresponding to the adjusting part **43** may be formed inside the seating part edge **54** so that the adjusting part **43** of the gap adjusting member **70** is inserted. In addition, an inclined surface **501** and a through-hole **502** may be formed in the center of the receiving space **500**, respectively.

(235) In addition, a support part **53** may be formed on the seating part **50**. The support part **53** may include a main support part **51** formed on a protruding rear surface of the seating part **50** and a sub support part **52** inside the main support part **51**.

(236) The main support part **51** includes a first main support part **511**, a second main support part **512**, a third main support part **513**, and a fourth main support part **514** and may be formed in a rectangular shape. In addition, the first main support part **511** has the same lowest height as the rear surface of the seating part edge **54**, and in the order of the second main support part **512**, the third main support part **513**, and the fourth main support part **514**, the height may be formed to increase stepwise.

(237) The sub support part **52** protrudes from an inner surface of the main support part **51** and may be disposed inside the receiving space **500**. The sub support part **52** may protrude vertically from each surface of the main support part **51** and may be formed of a plurality of protrusions.

(238) The sub support part **52** includes a first sub support part **521** protruding from an inner surface of the third main support part **513**, a second sub support part **522** protruding from an inner surface of the fourth main support part **514**, a third sub support part **523** protruding from the inner surface of the first main support part **511**, and a fourth sub support part **524** protruding from the inner surface of the second main support part **51**. In addition, the first sub support part **521** has the lowest height, and the height of the second sub support part **522**, the third sub support part **523**, and the fourth sub support part **524** can increase stepwise.

(239) In addition, a seating part guide **515** may be formed on an outer surface of the second sub

support part **522**.

(240) The gap adjusting member **70** having such a structure may be capable of adjusting the height by a simple manipulation without separate separation and disassembly manipulation.

(241) In a state where the door **20** is opened, when the gap adjusting member **70** is pulled rearward, the adjusting part **43** may be pulled out to the opened rear surface of the seating part **50** or further rearward therefrom. In this case, the elastic part **441** may be elastically deformed by being supported by the rear surface of the door **20**.

(242) In a state where the adjusting part **43** is moved outside the receiving space, the user may change the position of the adjusting part **43** by rotating the contact part **75**. In other words, by changing the disposition of the adjusting part **43** seated on the seating part **50** by the rotation of the contact part **75**, the support part **53** having a different height can be supported by the seating part **50**.

(243) When the contact part **75** is rotated so that the support part **53** can be disposed in a desired position, the gap adjusting member **70** may be rotated about the insertion protrusion **44** as a shaft. In addition, the selected height can be checked through the height indicating part **752**.

(244) When the gap adjusting member **70** is released after rotating the adjusting part **43** to a desired height, the adjusting part **43** may be inserted into the receiving space of the seating part **50** by an elastic force. In addition, the adjusting part **43** is supported by the support part **53** and at the same time can be inserted into the receiving space **500** so that the gap adjusting member **70** maintains a stable mounting state.

(245) In a state where the manipulation for adjusting the height of the gap adjusting member **70** is completed, the gap adjusting member **70** may be fixedly mounted to the seating part **50**. In addition, in a state where the gap adjusting member **70** is mounted on the seating part **50**, the contact part **75** and the seating part rim **54** have the same outer diameter, so that the gap adjusting member **70** and the circumferential surface of the seating part **50** does not protrude to the outside or recessed. In other words, in a state where the gap adjusting member **70** protrudes from the rear surface of the door **20**, the outer appearance thereof can be kept cleaner and more united.

(246) Meanwhile, a refrigerator according to an embodiment of the present disclosure may have various other embodiments in addition to the above-described embodiments. In a refrigerator according to another embodiment of the present disclosure, a pair of doors may be disposed side by side on both left and right sides, and a gap adjusting member may be provided on each of the left and right doors. In addition, the refrigerator according to another embodiment of the present disclosure differs only in the structure and disposition of the door, but all other components are the same, and the same components as in the above-described embodiment use the same reference numerals, and detailed descriptions thereof can be omitted or the illustration thereof can be omitted.

(247) FIG. **24** is a perspective view illustrating a refrigerator door according to another embodiment of the present disclosure viewed from the rear. In addition, FIG. **25** is a view schematically illustrating a state of adjusting a step between the doors of the refrigerator.

(248) As illustrated in the drawings, a pair of refrigerator doors **33** and **34** according to another embodiment of the present disclosure may be provided on both left and right sides. The pair of doors **33** and **34** may be referred to as a left door **34** and a right door **33**.

(249) The left door **34** and the right door **33** are disposed side by side to form a front outer appearance of the refrigerator **1**. In this case, the left door **34** and the right door **33** may be configured to open and close one storage space by rotation. In addition, the left door **34** and the right door **33** may be configured to open and close each storage space disposed on the left and right by rotation.

(250) The left door **34** and the right door **33** may have the same basic structure, and although not illustrated in detail, as in the above-described embodiment, the left door **34** and the right door **33** may include a front plate **21** forming the front surface, a door liner **25** forming a rear surface, an upper cap deco **22** forming an upper surface, and a lower cap deco **23** forming a lower surface.

(251) The door liner **25** may be provided with a gasket **26** that contacts the cabinet **10** and seals the storage space. In addition, an upper end corresponding to an area above the gasket **26** among the rear surfaces of the left door **34** and the right door **33** may be formed by the rear surface **222** of the upper cap deco **22**.

(252) A gap adjusting member **60** according to another embodiment of the present disclosure may be provided on rear surfaces of the left door **34** and the right door **33**, that is, the rear surface **222** of the upper cap deco **22**. The gap adjusting member **60** may protrude from the rear surface of the left door **34** and the right door **33**. In addition, the gap adjusting member **60** may be provided on a side far from the hinge mounting part **221** of the left and right sides of the upper cap deco **22**.

(253) In order to mount the gap adjusting member **60**, a seating part **50** may be formed on the rear surface of the upper cap deco **22**. The structure and shape of the gap adjusting member **60** and the seating part **50** are completely the same as those of the above-described embodiments, and thus a detailed description thereof will be omitted.

(254) The gap adjusting member **60** may be provided in both the left door **33** and **34** and the right door **33** and **34**, and if necessary, the gap adjusting member **60** can be provided only in any one of the left door **33** and **34** and the right door **33**, **34**.

(255) In addition, the gap adjusting member **60** may be mounted on the door liner **25** outside the gasket **26** when the door liner **25** extends outside the gasket **26**. In this case, the seating part **50** may be formed on the door liner **25**.

(256) Meanwhile, as illustrated in FIG. **25**, in a state where the left door and the right door are closed, the height of the front surface of the door may vary according to a state of mounting the door or a state of a load disposed on the door.

(257) In particular, in a structure in which the pair of doors **33** and **34** are disposed side by side with each other, the space between the left door **34** and the right door **33** may be very narrow. In this state, in a case where the left door **34** and the right door **33** have different heights from each other, a step between the pair of doors **33** and **34** may be very noticeable, and thus there is a problem of deteriorating the outer appearance.

(258) Therefore, as illustrated, in a case where the front surface of the right door **33** (as seen in FIG. **25**) protrudes more than the front of the left door **34** and a difference in height **H** occurs in the front and rear directions, the gap adjusting member **60** of the right door **33** is manipulated.

(259) The user opens the doors **33** and **34** to adjust the protruding height of the doors **33** and **34**, and then manipulates the gap adjusting member **60** mounted on the rear surface **222** of the right door **33** so that the protruding height of the gap adjusting member **60** may be lowered.

(260) The user pulls the gap adjusting member **60** to pull the adjusting part **43** out of the receiving space. Even in such a state, the gap adjusting member **60** maintains a state of being mounted on the right door **33**, but the elastic part **441** may be elastically deformed. After confirming that the user has rotated to the desired height through the height indicating part **615**, when the gap adjusting member **60** is released, the adjusting part **43** can be inserted into the inside of the receiving space **500** by the elastic force of the elastic part **441**.

(261) By adjusting the protruding height of the gap adjusting member **60** as described above, the distance **D** between the rear surface of the right door **33** and the front surface of the cabinet **10** can be adjusted to become narrower. Accordingly, the front surfaces of the doors **33** and **34** are lowered by the adjusted height, and finally, the front surfaces of the doors **33** and **34** can be positioned on the same plane as the front surfaces of the left door **34**.

(262) Meanwhile, a refrigerator according to an embodiment of the present disclosure may have various other embodiments in addition to the above-described embodiments. In a refrigerator according to another embodiment of the present disclosure, a gap adjusting member may be provided on a top cover. In addition, in the refrigerator according to another embodiment of the present disclosure, only the position of the gap adjusting member is different, but all other configurations are the same, and the same configurations as in the above-described embodiment

use the same reference numerals, and detailed descriptions thereof can be omitted or the illustration thereof can be omitted.

(263) FIG. **26** is an exploded perspective view illustrating a state where a refrigerator door is separated according to another embodiment of the present disclosure.

(264) As illustrated in the drawings, a refrigerator **1** according to another embodiment of the present disclosure may include a cabinet **10** in which a storage space is formed, and a door **20** for opening and closing the storage space.

(265) An upper hinge **201** may be provided on an upper surface of the cabinet **10**, and the upper hinge **201** may be connected to a hinge mounting part **221** at one end of the door **20**. Accordingly, the door **20** may open and close the storage space by rotating about the upper hinge **201** as a shaft.

(266) In addition, a top cover **11** may be provided on an upper front surface of the cabinet **10**. The top cover **11** may extend from an upper surface of the cabinet **10** to a position corresponding to a height of an upper end of the door **20**. Accordingly, the top cover **11** may be positioned at a position facing each other with the rear surface of the door **20**.

(267) A front surface of the top cover **11** may be formed in a planar shape, and a display **112** may be disposed to display an operating state of the refrigerator **1**. In addition, if necessary, an electric component or a PCB may be further provided inside the top cover **11**.

(268) Meanwhile, the gap adjusting member **60** may be provided on one side of the top cover **11**. The gap adjusting member **60** may be disposed on one side of the left and right sides of the top cover **11** and the other side far from the side on which the upper hinge **201** is disposed. Further, the gap adjusting member **60** may protrude forward, and thus, when the door **20** is closed, the door **20** contacts the rear surface of the door **20** to support the door **20**.

(269) The gap adjusting member **60** may be positioned higher than the upper surface of the cabinet **10**, and thus, the gap adjusting member **60** is positioned higher than the gasket **26** on the rear surface of the door **20**, so that the door **20** does not interfere with the gasket **26** when the door **20** is opened or closed.

(270) Meanwhile, in order to mount the gap adjusting member **60**, the seating part **50** may be formed on the front surface of the top cover **11**. Since the shape of the seating part **50** and the shape and coupling structure of the gap adjusting member **60** are exactly the same as those of the above-described embodiment, reference may be made to the above-described embodiment, and a detailed description thereof will be omitted.

(271) The protruding height of the door **20** can be adjusted by pulling and rotating the gap adjusting member **60** as in the above-described embodiment, and when the gap adjusting member **60** is placed at a desired height, the gap adjusting member **60** can have a structure that is mounted on the seating part **50** again.

(272) Accordingly, it is possible to adjust the step of the door **20** by mounting the gap adjusting member **60** on the top cover **11** without changing the structure of the door **20**.

(273) Of course, in a case where the height of the cabinet **10** is higher so that the gap adjusting member **60** can be disposed, the gap adjusting member **60** may be installed on the front surface of the cabinet **10**.

(274) Meanwhile, a refrigerator according to an embodiment of the present disclosure may have various other embodiments in addition to the above-described embodiments. The refrigerator according to another embodiment of the present disclosure may have a structure in which a separate elastic part is not formed in the gap adjusting member, and the gap adjusting member can be pulled out and returned by the elastic deformation of the gap adjusting member itself. In addition, some structures of the gap adjusting member and the structure of the seating part according to another embodiment of the present disclosure may be the same as those of the above-described embodiment, and reference numerals not illustrated may refer to the above-described embodiments. Accordingly, the same configurations as those of the above-described embodiment use the same reference numerals, and detailed descriptions thereof may be omitted or the

illustration thereof may be omitted.

(275) FIG. 27 is a perspective view illustrating a refrigerator according to another embodiment of the present disclosure. In addition, FIG. 28 is a partial perspective view illustrating an upper portion of the cabinet of the refrigerator.

(276) As illustrated in the drawings, a refrigerator 1'' according to another embodiment of the present disclosure may be installed alone like a general refrigerator. The refrigerator 1'' may be externally formed by a cabinet 10' forming a storage space and doors 36 and 37 that open and close the front surface of the cabinet 10'.

(277) For example, the storage space of the cabinet 10' may be divided up and down and may form an upper storage space 12 and a lower storage space. In addition, the upper storage space 12 may be opened and closed by a pair of upper doors 36. The upper doors 36a and 36b are disposed side by side on both left and right sides and may be configured to open and close the upper storage space 12 by rotation. In addition, the lower storage space may be opened and closed by a pair of lower doors 37a and 37b. The lower door 37 is also disposed side by side on both left and right sides and may be configured to open and close the lower storage space by rotation.

(278) As another example, the storage space of the cabinet 10' may be divided left and right. In this case, the doors may be disposed side by side on both left and right sides to open and close the storage space on the left and the storage space on the right.

(279) Meanwhile, a top cover 13 may be provided on an upper surface of the cabinet 10'. The top cover 13 may be located at a front end of the cabinet 10', and hinge cover parts 131 for receiving a hinge device coupled to an upper end of the upper door 36 may be formed at both side ends. Accordingly, the top cover 13 may be referred to as a hinge cover.

(280) The top cover 13 may extend from a left end to a right end of the upper surface of the cabinet 10' and may protrude further upward than the upper surface of the cabinet 10'. In addition, the top cover 13 has an opened lower surface to receive the hinge device therein. In addition, a PCB for controlling the operation of the refrigerator 1'' may be provided inside the top cover 13 as needed, and a door switch for sensing opening/closing of the upper door 36 may be provided.

(281) In addition, a display 14 may be disposed in the center of the top cover 13. The display 14 may be mounted inside the top cover 13 and may be exposed to the front through an opening 132 in front surface of the top cover 13.

(282) The height of the top cover 13 may be equal to or slightly lower than the height of the upper end of the upper door 36. Accordingly, in a state where the upper door 36 is closed, the front surface of the top cover 13 is not exposed to the front, and the display 14 is also not exposed.

(283) When the upper door 36 is opened, the display 14 may be exposed forward, and operation information of the refrigerator 1'' may be displayed through the display 14. Of course, the display 14 not only displays information, but also includes a manipulation part that a user directly manipulates and may allow a user's manipulation input. For example, the display 14 may be a touch display capable of inputting a touch screen operation.

(284) The display 14 may be disposed in a central area of the front surface of the top cover 13 and may be disposed at least in an area between the gap adjusting members 80 provided in the upper door 36. Therefore, when the upper door 36 is closed, the display may not come into contact with the gap adjusting member 80 protruding from the rear side of the upper door 36. In other words, the gap adjusting member 80 to be described in detail below may contact the front surface of the top cover 13 that is outside the display 14 when the upper door 36 is closed.

(285) Hereinafter, the structure of the gap adjusting member 80 will be described in more detail with reference to the drawings.

(286) FIG. 29 is a rear view illustrating a refrigerator door according to another embodiment of the present disclosure. In addition, FIG. 30 is a perspective view illustrating a gap adjusting member according to another embodiment of the present disclosure. In addition, FIG. 31 is a cut-away perspective view taken along line XXXI-XXXI' of FIG. 29.

(287) As illustrated in the drawing, a pair of the upper doors **36** may be disposed on both left and right sides and may be composed of an upper left door **36a** and an upper right door **36b**, respectively. The upper left door **36a** and the upper right door **36b** may have the same basic structure and size. Of course, if necessary, the upper left door **36a** and the upper right door **36b** may have different configurations. For example, a door on one side may have a door in door (DID) structure disposed in two layers and may be configured in the form of a transparent door capable of seeing the inside. However, the size and shape will be the same when viewed from the front, and the basic structure will also be the same. Accordingly, hereinafter, both the upper left door **36a** and the upper right door **36b** may be referred to as a door or an upper door **36**.

(288) In detail, the upper door **36** may include an out plate **361** forming an outer surface exposed to the outside of the refrigerator, a door liner **362** forming an inner surface facing the inside of the refrigerator, an upper cap deco **363** that connects the out plate **361** and the upper end of the door liner **362** and forms the upper surface of the upper door **36**, and a lower cap deco **364** that connects the outer plate **361** and the lower end of the door liner **362** and connects the lower surface of the upper door **36**. In addition, the outer plate **361**, the door liner **362**, the upper cap deco **363**, and the lower cap deco **364** may form a space in which the insulating material **400** can be filled.

(289) In addition, a gasket **365** in contact with the front surface of the cabinet **10'** may be further provided along the circumference of the door liner **362**. In addition, although not illustrated in detail, the upper door **36** on one side may be provided with an opening and closing member for opening and closing a space between the pair of upper doors **36**, and a plurality of door baskets may be provided on the rear surface of the door liner **362**.

(290) Meanwhile, the upper cap deco **363** may form a part of the rear surface of the upper door **36**. In addition, a gap adjusting member **80** may be provided on the rear surface of the upper cap deco **363**. The gap adjusting member **80** protrudes from the rear surface of the upper door **36** and can adjust the height protruding from the rear surface of the upper door **36** by a user's manipulation. Of course, the gap adjusting member **80** may be provided in the lower cap deco **364** as necessary, and may be configured to be in contact with one side of the cabinet **10'**.

(291) Therefore, in a state where the upper door **36** is closed, the distance between the rear surface of the upper door **36** and the cabinet **10'** can be adjusted, and as a result, the height of protrusion of the front surface of the upper door **36** in the front and rear direction can be adjusted. In addition, by adjusting the protruding height of the front surface of the upper door **36** in the front and rear direction, it is possible to eliminate a height difference in the front and rear direction between the pair of upper doors **36**.

(292) The gap adjusting member **80** may be located at one side away from both left and right side ends of the upper door **36**. The gap adjusting member **80** may be positioned further outside the display **14** in a closed state. Accordingly, the gap adjusting member **80** may be disposed at a position further spaced apart than half the length of the left and right sides of the display **14** from one end of the upper door **36** far from the rotation axis of the upper door **36** among the left and right ends of the upper door **36**.

(293) In addition, the gap adjusting member **80** is positioned as close to the display **14** as possible within a range that does not interfere with the display **14** in a state where the upper door **36** is closed, so that the height of the upper door **36** in the front and rear direction can be easily adjusted.

(294) The gap adjusting member **80** may protrude from the rear surface of the upper cap deco **363**. In addition, in order to mount the gap adjusting member **80**, a seating part **50** may be formed on the rear surface of the upper cap deco **363**. The structure and shape of the seating part **50** may be completely the same as in the above-described embodiment.

(295) The seating part **50** may have a polygonal cross-sectional shape and may be formed to have an opened rear surface. In addition, a receiving space **500** may be formed inside the seating part **50**, and an inclined surface **501** and a through-hole **502** may be formed in the center.

(296) In addition, a support part **53** may be formed on the seating part **50**. The support part **53** may

include a main support part **51** forming an outer circumference shape of the seating part **50** and a sub support part **52** inside the main support part **51**.

(297) The main support part **51** includes a first main support part **511**, a second main support part **512**, a third main support part **513**, and a fourth main support part **514** and may be formed in a rectangular shape. In addition, the first main support part **511** has the lowest height, and the height of the second main support part **512**, the third main support part **513**, and the fourth main support part **514** can be formed to increase stepwise in order of the second main support part **512**, the third main support part **513**, and the fourth main support part **514**.

(298) The sub support part **52** protrudes from an inner surface of the main support part **51** and may be disposed inside the receiving space **500**. The sub support part **52** may protrude vertically from each surface of the main support part **51** and may be formed of a plurality of protrusions.

(299) The sub support part **52** may include a first sub support part **521** protruding from an inner surface of the third main support part **513**, a second sub support part **522** protruding from an inner surface of the fourth main support part **514**, a third sub support part **523** protruding from the inner surface of the first main support part **511**, and a fourth sub support part **524** protruding from the inner surface of the second main support part **51**. In addition, the first sub support part **521** has the lowest height, and the height of the second sub support part **522**, the third sub support part **523**, and the fourth sub support part **524** can be formed to increase stepwise in order of the second sub support part **522**, the third sub support part **523**, and the fourth sub support part **524**.

(300) Meanwhile, the gap adjusting member **80** may include a contact part **85** forming a rear surface in contact with the cabinet **10**, an adjusting part **43**, and an insertion protrusion **44** formed on the front surface of the contact part **85**.

(301) The gap adjusting member **80** is formed of a rubber or silicone material and is elastically deformed when the gap adjusting member **80** is pulled to provide an elastic force. In addition, the gap adjusting member **80** may be formed of an elastic material to mitigate an impact generated when contacting the cabinet **10**.

(302) An adjusting part **43** and the insertion protrusion **44** may be formed on the front surface of the contact part **85**. The shape and structure of the adjusting part **43** and the insertion protrusion **44** may be the same as those of the above-described embodiment, and reference numerals not illustrated may refer to the above-described embodiments.

(303) The adjusting part **43** may include a main adjusting part **41** protruding from a position corresponding to the main support part **51** and a sub adjusting part **42** protruding from a position corresponding to the sub support part **52**.

(304) The main adjusting part **41** is disposed along the circumference of the sub adjusting part **42**, in order of height, the first main adjusting part **411**, the second main adjusting part **412**, the third main adjusting part **413**, and a fourth main adjusting part **414**. In addition, the sub adjusting part **42** is located inside the main adjusting part **41** and may include the first sub adjusting part **421** inside the third main adjusting part **413**, and a second sub adjusting part **422** inside the fourth main adjusting part **414**, a third sub adjusting part **423** inside the first main control part **411**, and a fourth sub adjusting part **424** inside the second main control part **412**.

(305) The structure of the adjusting part **43** and the structure of the seating part **50** in contact with the adjusting part **43** may be the same as those of the above-described embodiment, and reference numerals not illustrated refer to the above-described embodiments. Further, the position of the adjusting part **43** in contact with the seating part **50** is changed by the rotation of the adjusting part **43**, so that the height of the gap adjusting member **80** can be adjusted.

(306) The contact part **85** forms a rear surface in contact with the cabinet **10** and may be formed in a planar shape. In addition, a height indicating part **852** may be formed on the contact part **85**. The height indicating part **852** may be formed in the shape of a recessed groove, and the height thereof may be displayed by the number of recessed grooves.

(307) In this case, the height indicating part **852** may be arranged to rotate at 90° angle intervals

based on the center of the contact part **85**. In addition, the height indicating part **852** may be positioned at a position corresponding to the position of the adjusting part **43** to display the height of the corresponding adjusting part **43** to the user.

(308) Meanwhile, the insertion protrusion **44** may extend forward from the center of the sub adjusting part **42** and may be inserted into the door **20** through the through-hole **502**.

(309) The insertion protrusion **44** may have a circular cross section so as to be a rotation shaft of the gap adjusting member **80** and may be formed to have a predetermined diameter. In addition, a locking part **442** may be formed in the insertion protrusion **44**.

(310) The locking part **442** is locked by the through-hole **502** in a state where the gap adjusting member **80** is mounted, and even when the step adjustment member **80** is pulled, the step adjustment member **80** can be prevented from being separated from the seating part **50**.

(311) In detail, the locking part **442** (restrain part) may protrude from an outer surface of the insertion protrusion **44** corresponding to the position of the through-hole **502** in a state where the gap adjusting member **80** is mounted. The locking part **442** may protrude along an outer circumference of the insertion protrusion **44**, and the locking part **442** may be formed to have an inclination that decreases in protrusion height toward the rear.

(312) When the gap adjusting member **80** is first mounted on the upper cap deco **363**, the inclined circumferential surface of the locking portion **442** is moved forward in contact with the inclined surface **501** and passes through the through-hole **502** while being elastically deformed. After passing through the through-hole **502**, the front end of the locking part **442** is restrained while supporting the circumferential surface of the through-hole **502** and thus it is prevented that the locking part **442** passes through the through-hole **502** and falls out to the rear. The gap adjusting member **80** may be mounted in a state where the locking portion **442** passes through the through-hole **502** and is locked, and the gap adjusting member **80** can be manipulated to adjust the height of the gap adjusting member **80**.

(313) Meanwhile, the insertion protrusion **44** may include a first protrusion **443** and a second protrusion **444**. In detail, the insertion protrusion **44** may include a first protrusion part **443** forming a space between the locking part **442** and the adjusting part **43** based on the locking part **442** and a second protrusion part **444** that forms up to the extended end portion of the insertion protrusion **44** from the locking part **442**. The first protrusion part **443** and the second protrusion part **444** may be formed to have the same thickness T1, that is, the same outer diameter, in a state where no external force is applied.

(314) The first protrusion part **443** is positioned in an inner region of the seating part **50** in a state where the gap adjusting member **80** is mounted and is positioned in an outer region of the upper cap deco **363**. In addition, when pulled rearward for height adjustment of the gap adjusting member **80**, the first protrusion part **443** may be extended while being elastically deformed, and the first protrusion part **443** may have a thickness T2 which becomes relatively thinner than the thickness T1.

(315) In addition, the first protrusion part **443** may be twisted by a rotation manipulation of the gap adjusting member **80** in an extended state. Further, when the gap adjusting member **80** is pulled and released, it may be seated on the seating part **50** again by the restoring force of the first protrusion part **443**.

(316) The second protrusion part **444** may be located in an area in front of the seating part **50** in a state where the gap adjusting member **80** is mounted and may be located in an inner area of the upper cap deco **363**. The second protrusion part **444** may be buried in the space of the insulating material **400** in a case where the insulating material **400** is filled inside the upper cap deco **363**.

(317) In addition, the second protrusion part **444** may maintain its initial thickness and state even when manipulating the height adjustment of the gap adjusting member **80**. In other words, even if the gap adjusting member **80** is pulled backward, the position of the second protrusion part **444** is not changed by the locking part **442**, and thus the second protrusion part can maintain the state at

the time of the first insertion and installation.

(318) The gap adjusting member **80** having such a structure may be able to adjust the height by simple manipulation without separate separation and disassembly manipulation.

(319) FIG. **32** is a cut-away perspective view illustrating a manipulation state of the gap adjusting member. In addition, FIG. **33** is a partial perspective view illustrating a state where the door of the refrigerator is closed.

(320) As illustrated in the drawing, in order to adjust the front protrusion height of the front of the upper door **36** by manipulating the gap adjusting member **80**, first, hold the contact part **85** and pull the gap adjusting member **80** forward.

(321) By such a manipulation, the first protrusion part **443** of the insertion protrusion **44** may extend while being elastically deformed, and as illustrated in FIG. **32**, the adjusting part **43** can be pulled out to the opened rear surface of the seating part **50**. At this time, the first protrusion part **443** has an elastic restoring force by tension. Further, the second protrusion **444** is maintained in a state of being inserted from the inside of the upper cap deco **363** by the locking part **442**.

(322) In a state where the adjusting part **43** is moved to the outside of the receiving space **500**, the user may change the position of the adjusting part **43** by rotating the contact part **85**. In other words, it is possible to change the disposition of the adjusting part **43** seated on the seating part **50** by rotating the contact part **85**, and the support part **53** having a different height can be supported by the seating part **50**.

(323) When the contact part **85** is rotated so that the support part **53** can be disposed in a desired position, the gap adjusting member **80** may be rotated about the insertion protrusion **44** as a shaft. In addition, the selected height can be checked through the height indicating part **852**.

(324) When the gap adjusting member **80** is released after rotating the adjusting part **43** to a desired height, the adjusting part **43** as illustrated in FIG. **31** due to the elastic restoring force of the first protrusion part **443** may be inserted into the receiving space **500** of the seating part **50**. In addition, the adjusting part **43** is supported by the support part **53** and at the same time can be inserted into the receiving space **500** so that the gap adjusting member **80** can maintain a stable mounting state.

(325) As illustrated in FIG. **33**, in a state where the upper door is closed, the gap adjusting member provided in the upper left door and the upper right door may contact the cabinet or the top cover, respectively, and may contact the cabinet and the top cover at the same time.

(326) In addition, the position of the gap adjusting member **80** may be located in the left and right sides more than the display **14**, so even if the upper door **36** is strongly closed or repeatedly opened and closed, a direct impact is transmitted to the display **14** so that it is not damaged or a mark is not left on the display **14**.

(327) It is possible to manipulate the protruding height of the gap adjusting member **80** through the manipulation of the gap adjusting member **80**, so that the front surface of the pair of upper doors **36** can be adjusted to have the same height in the front and rear directions. In other words, by a user, the pair of upper doors **36** may be adjusted so that a step does not occur with each other.

(328) Of course, the same structure may also be applied to the lower door **37**, and it will be possible to adjust the step between the pair of lower doors **37** and to adjust the step between the upper door **36** and the lower door **37** disposed vertically.

(329) In the refrigerator according to the proposed embodiment, the following effects can be expected.

(330) In the refrigerator according to the embodiment of the present disclosure, the protruding height of the gap adjusting member can be adjusted by manipulating the gap adjusting member while the gap adjusting member is mounted. Therefore, it is possible to immediately and easily adjust the front and rear protruding height of the door, without the need for additional operations, such as separation and mounting.

(331) Therefore, There is an advantage that the height of the front surface of the door in the front and rear direction may be the same as the height of the door of the neighboring refrigerator in the

front and rear direction or the height of other adjacent doors in the same refrigerator in the front and rear direction, and, due to the built-in mounting, it can be adjusted to have the same height as neighboring furniture or walls in the front and rear direction and thus the installation outer appearance thereof is improved. In particular, in a case where a plurality of home appliances or refrigerators are built-in, these advantages can be maximized.

(332) In addition, there is an advantage that, when the gap adjusting member is rotated to a desired position after pulling the step adjusting part for height adjusting, the height adjusting member is moved to the mounted state by the elastic force of the elastic part, so that the height adjusting operation is very simple.

(333) In addition, since there is no need to disassemble or separate a separate configuration, there is no need for management of the disassembled configuration during the height adjusting operation, and there is no fear of loss or damage of the configuration, so there is an advantage that maintenance is easy and durability is guaranteed.

(334) In addition, since the elastic part has a structure integrally formed with the gap adjusting member made of an elastic material, the gap adjusting member can be completed in a single configuration, and the structure thereof is very simple. Therefore, there is an advantage that productivity thereof can increase and manufacturing cost thereof can be reduced.

(335) In addition, the gap adjusting member may be formed of a material in which an insertion protrusion serving as a rotation shaft of the gap adjusting member is elastically deformable. Therefore, when the gap adjusting member is pulled for height adjusting, rotated to a desired position, and then gets higher, the contact part and the adjusting part are pulled out of the seating part in a state where the mounting of the gap adjusting member is maintained by the elastic deformation of the insertion protrusion itself. In such a situation, when the contact part is released after the rotation manipulation of the gap adjusting member, the adjusting part is returned to a state where the adjusting part is seated in the seating part by the elastic restoring force of the insertion part, so that the height adjusting operation is very simple.

(336) In addition, since the height indicating part indicating the height to be manipulated is formed on the contact part, the user can check the height to be adjusted through the height indicating part, and thus, there is an advantage of enabling a more accurate and easy height adjusting operation.

(337) In addition, an adjusting member guide is formed in the gap adjusting member, and a seating part guide is formed in the seating part, so that it is easy to align and mount the gap adjusting member and the seating part, and thus misassembly of the gap adjusting member is prevented, and mounting at an accurate position is ensured, thereby ensuring manipulation reliability.

(338) In addition, the adjusting part of the gap adjusting member is formed in a polygonal shape corresponding to the inner shape of the seating part, and when the gap adjusting member is mounted, the adjusting part may be inserted into the seating part. In this case, the adjusting part has a different height along each side of the adjusting part, and thus, when the gap adjusting member is inserted, accurate height adjustment is ensured and misassembly is prevented, thereby ensuring reliability of manipulation.

(339) In addition, in a case where the display is disposed above the cabinet with which the gap adjusting member is in contact, the position of the gap adjusting member may be disposed at a position not in contact with the display, and according to this, even if the door is strongly closed or a repetitive opening and closing operation is performed, it is possible to prevent damage to the display, and there is an advantage in that it is possible to prevent the outer appearance thereof from being deteriorated due to traces caused by the contact with the gap adjusting member.

Claims

1. A refrigerator comprising: a cabinet that defines a storage space therein; a door configured to open and close at least a portion of the storage space; a seating part that protrudes from a rear

surface of the door; and a gap adjusting member that is movably inserted into the seating part, wherein the gap adjusting member comprises: a contact part in contact with a front of the cabinet, an insertion protrusion that extends from the contact part and passes through a through-hole defined at the seating part, a restrain part that protrudes from the insertion protrusion and that is configured to be restrained in the through-hole based on movement of the gap adjustment member, and a plurality of adjusting parts that protrude from the contact part and are arranged in a rotational direction around the insertion protrusion, the plurality of adjusting parts having different heights from one another, wherein a gap between the door and the cabinet is configured to be adjusted by a rearward movement and rotation of the contact part to thereby select one of the different heights of the plurality of adjusting parts in contact with the seating part, and wherein a plurality of support parts is disposed on the seating part, protrudes toward the cabinet at different protrusion heights, and is configured to support the gap adjusting member.

2. The refrigerator of claim 1, wherein the gap adjusting member is at least partially made of a deformable material.

3. The refrigerator of claim 2, wherein the insertion protrusion is made of a deformable material, and wherein, based on the contact part moving rearward, the insertion protrusion extends elastically while being fixed to the through-hole.

4. The refrigerator of claim 2, wherein the restrain part protrudes along an outer circumference of the insertion protrusion, and wherein the restrain part has an inclination such that a protrusion height of the restrain part decreases toward a rear of the contact part.

5. The refrigerator of claim 4, wherein the restrain part is made of a deformable material, configured to provide elastic force so that the gap adjusting member is returned to an inserted state in the seating part after manipulation.

6. The refrigerator of claim 5, wherein the restrain part is formed to be further away from an outer surface of the insertion protrusion as it extends rearward.

7. The refrigerator of claim 5, wherein the contact part is moved rearward, and the restrain part supported by a circumferential surface of the through-hole is elastically deformed.

8. The refrigerator of claim 4, wherein a diameter of the restrain part is greater than a diameter of the through-hole.

9. The refrigerator of claim 2, wherein an inclined surface is formed around the through-hole such that an inner diameter becomes narrower from a front to a rear of the through-hole, and wherein the through-hole is formed at a front end of the inclined surface.

10. The refrigerator of claim 9, wherein the restrain part is formed to have an inclination corresponding to the inclined surface.

11. The refrigerator of claim 1, wherein the gap adjusting member is configured to select contact surfaces between the plurality of adjusting parts so that a height of the gap adjusting member is adjusted by a combination of the contact surfaces of the plurality of adjusting parts and the plurality of support parts.

12. The refrigerator of claim 1, wherein the plurality of adjusting parts comprise: a main adjusting part that is disposed along each side of the contact part; and a sub adjusting part that is disposed inside the main adjusting part and further protrudes toward the seating part relative to the main adjusting part, the sub adjusting part being disposed along each side of the main adjusting part, and wherein the main adjusting part and the sub adjusting part define a plurality of protrusion height differences along sides of the contact part, respectively, each side of the contact part having one of the plurality of protrusion height differences.

13. The refrigerator of claim 12, wherein the main adjusting part and the sub adjusting part are located opposite to each other with respect to a center of the contact part, and wherein the gap adjusting member is configured to rotate by 180° about the center of the contact part to thereby switch positions of the main adjusting part and the sub adjusting part.

14. The refrigerator of claim 12, wherein a cross-sectional area of the sub adjusting part is equal to

a portion of an inside area of a receiving space of the seating part, and the sub adjusting part is configured to be inserted into the receiving space.

15. The refrigerator of claim 12, wherein the plurality of support parts comprises: a main support part that defines a circumferential surface of the seating part and supports the main adjusting part, and a sub support part that is disposed inside the main support part and supports the sub adjusting part, and wherein the main support part and the sub support part define a plurality of support steps corresponding to the plurality of protrusion height differences, respectively.

16. The refrigerator of claim 1, wherein the gap adjusting member is configured to rotate about the insertion protrusion.

17. The refrigerator of claim 1, wherein the plurality of adjusting parts are arranged in a protrusion height order.

18. The refrigerator of claim 1, further comprising an upper cap deco that is disposed at an upper end of the door and defines an upper surface and the rear surface of the door, wherein the seating part is disposed at the upper cap deco.

19. The refrigerator of claim 1, further comprising a top cover that is disposed at an upper surface of the cabinet and protrudes upward relative to the upper surface of the cabinet, the top cover facing the door, wherein the contact part is configured to be in contact with the top cover based on the door being closed.
